

# **TABLES, GRAPHS AND SUMMARY MEASURES IN THE ANALYSIS OF TWO VARIABLES**

# The study of two variables [READING]

In many applications, one is interested to study two variables and their relationship:

- ▶ Is there a relationship between consumers' attention/interest in healthy eating and their age?
- ▶ Are there differences in the response times to complaint tweets from companies incorporating social media into their marketing strategy?
- ▶ Is there a relationship between performance in the Bachelor's degree and performance in the Master's degree? Is it possible to use the possible relationship to select candidates?

# The joint analysis of two variables [READING]

Also when studying two variables, it is important to effectively organise the collected data, and to use tables and graphs to unveil data's most salient features.

Different tools can be used depending on the **type of data** and on the number of values taken by the variables.

In particular, we consider the joint analysis of two variables

- ▶ both qualitative (or, more generally, with a limited number of categories)
- ▶ one qualitative (or discrete with a limited number of distinct values) and one numerical (typically continuous)
- ▶ both numerical (typically both continuous)

## **Joint analysis of two variables**

**Qualitative variables (or discrete taking few distinct values)**

# Tables and graphs for qualitative variables

Qualitative (or discrete) variables typically take a small number of categories (distinct values).

To organise the data in this case, we consider the **joint frequency distribution**, which describes:

- ▶ The distinct **pairs of categories** observed on the two variables,
- ▶ The **relevance** (frequency or percentage) of the observed pairs of categories, i.e:
  - ▶ their joint **absolute** frequencies (number of cases for each pair of categories);
  - ▶ Their joint **relative** frequencies (proportion of cases for each pair of categories, obtained by dividing the relative frequencies by the total number of cases) and/or the corresponding **percentages**.

The joint frequency distribution is organised in a **contingency table (or cross table)** displaying frequencies of all the possible pairs of values.

# Qualitative variables: cross tabulation

## Cross-tab (absolute frequencies)

|                              |         |                              |          |     |          |
|------------------------------|---------|------------------------------|----------|-----|----------|
|                              |         | Distinct values taken by $Y$ |          |     |          |
|                              |         | $y_1^*$                      | $y_2^*$  | ... | $y_J^*$  |
| Distinct values taken by $X$ | $x_1^*$ | $f_{11}$                     | $f_{12}$ | ... | $f_{1J}$ |
|                              | $x_2^*$ | $f_{21}$                     | $f_{22}$ | ... | $f_{2J}$ |
|                              | ...     | ...                          | ...      | ... | ...      |
|                              | $x_K^*$ | $f_{K1}$                     | $f_{K2}$ | ... | $f_{KJ}$ |

Each cell reports the frequency (in this case, the absolute frequency) characterising each pair of values

For the sake of simplicity, we will always denote by  $Y$  the variable on the columns and by  $X$  the variable on the rows of the table

# Qualitative variables: cross tabulation

## Cross-tab: absolute frequencies

|                              |         |                              |          |     |          |       |
|------------------------------|---------|------------------------------|----------|-----|----------|-------|
|                              |         | Distinct values taken by $Y$ |          |     |          | Total |
|                              |         | $y_1^*$                      | $y_2^*$  | ... | $y_J^*$  |       |
| Distinct values taken by $X$ | $x_1^*$ | $f_{11}$                     | $f_{12}$ | ... | $f_{1J}$ | $R_1$ |
|                              | $x_2^*$ | $f_{21}$                     | $f_{22}$ | ... | $f_{2J}$ | $R_2$ |
|                              | ...     | ...                          | ...      | ... | ...      | ...   |
|                              | $x_K^*$ | $f_{K1}$                     | $f_{K2}$ | ... | $f_{KJ}$ | $R_K$ |
|                              | Total   | $C_1$                        | $C_2$    | ... | $C_J$    | $n$   |

Column-wise sums of joint frequencies: frequencies of the distinct values taken by the variable on the columns ( $Y$ ). This is the frequency distribution of  $Y$ , also said **marginal frequency distribution** of  $Y$  in this context

Row-wise sums of joint frequencies: frequencies of the distinct values taken by the variable on the rows ( $X$ ). Thus, this is the frequency distribution of  $X$ , also said **marginal frequency distribution** of  $X$  in this context

The sum of all the **absolute** joint frequencies is the number of cases (here we consider a **sample** of size  $n$ ).

# Qualitative variables: double-entry table

Example: On a sample of 1224 individuals born in Italy or who lived in Italy for at least 20 years, information was collected on the attention/interest in healthy eating, as well as on some socio-demographic variables, including age (in classes).

| <b><i>Age/Interest</i></b> | Low        | Medium     | High       | Tot         |
|----------------------------|------------|------------|------------|-------------|
| 18-30                      | 183        | 272        | 49         | 504         |
| 31-45                      | 97         | 168        | 68         | 333         |
| 46-60                      | 65         | 214        | 108        | 387         |
| <b>Tot</b>                 | <b>345</b> | <b>654</b> | <b>225</b> | <b>1224</b> |

**Absolute joint frequencies**

| <b><i>Age/Interest</i></b> | Low         | Medium      | High        | Tot         |
|----------------------------|-------------|-------------|-------------|-------------|
| 18-30                      | 0.15        | 0.22        | 0.04        | 0.41        |
| 31-45                      | 0.08        | 0.14        | 0.06        | 0.28        |
| 46-60                      | 0.05        | 0.17        | 0.09        | 0.31        |
| <b>Tot</b>                 | <b>0.28</b> | <b>0.53</b> | <b>0.19</b> | <b>1.00</b> |

**Relative joint frequencies  
(Absolute freq. / 1224)**



# Qualitative variables: bar charts

To graphically represent a cross table, **bar charts** are typically used:

- ▶ **Side by side bar plots:** for each pair of categories one bar is built, having height proportional to the frequency of the pair. The bars are coloured according to the values of one variable and are grouped according to the values of the other.
- ▶ **Stacked bar plots:** to each value of one of the two variables a bar is associated, built by stacking one bar for each value of the second variable. The height of the stacked bars is proportional to the frequency of the corresponding pair of values

# Qualitative variables: bar charts

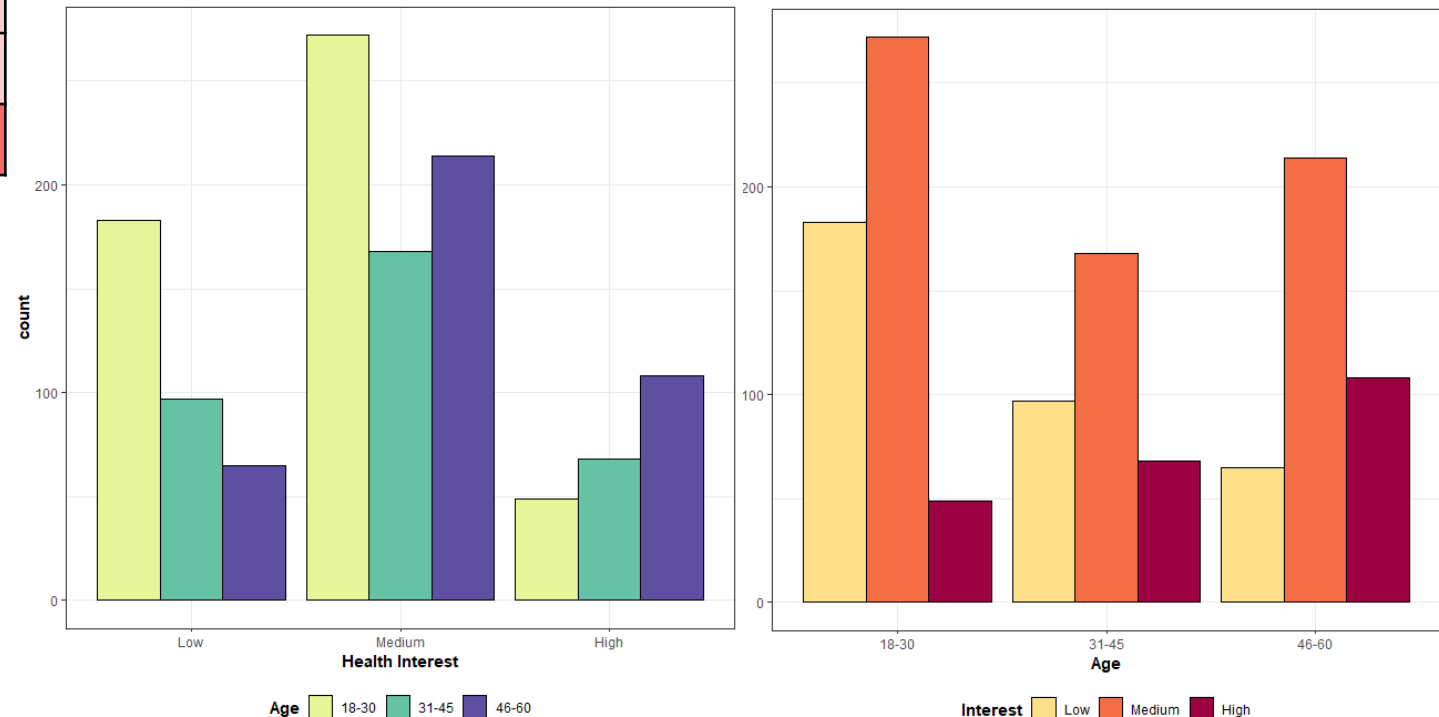
Example: Age and interest in healthy eating (sample of 1224 individuals)

| <i>Interest</i><br>Age | Low | Medium | High | Tot  |
|------------------------|-----|--------|------|------|
| 18-30                  | 183 | 272    | 49   | 504  |
| 31-45                  | 97  | 168    | 68   | 333  |
| 46-60                  | 65  | 214    | 108  | 387  |
| Tot                    | 345 | 654    | 225  | 1224 |

Side-by-side bar diagrams

The two diagrams show the **same bars**,  
**grouped by Age** or by **Interest** in  
healthy eating

They allow to easily identify the most  
relevant combinations of categories  
and to appreciate their relevance in  
relative terms (the **appearance** of the  
graphs **does not change** when  
considering relative frequencies)



# Qualitative variables: bar charts

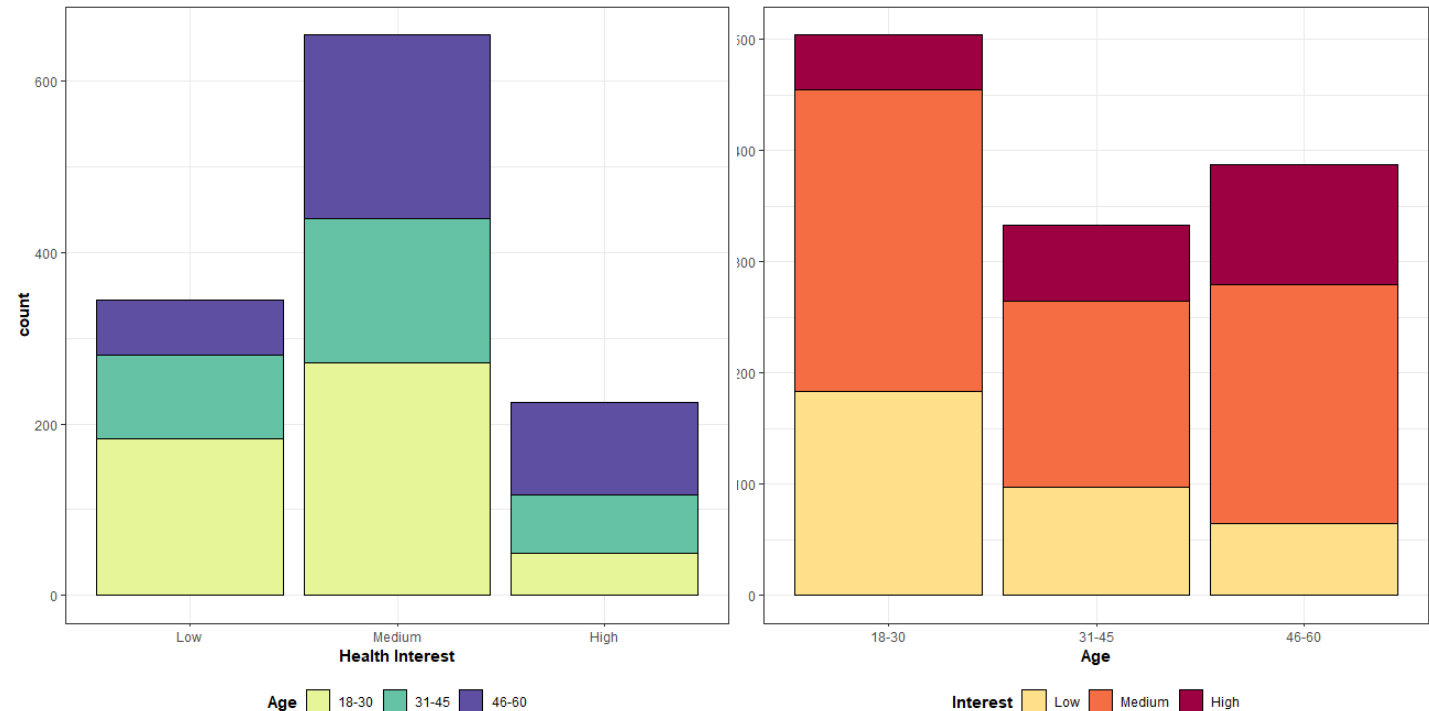
Example: Age and interest in healthy eating (sample of 1224 individuals)

| <i>Interest</i><br>Age | Low | Medium | High | Tot  |
|------------------------|-----|--------|------|------|
| 18-30                  | 183 | 272    | 49   | 504  |
| 31-45                  | 97  | 168    | 68   | 333  |
| 46-60                  | 65  | 214    | 108  | 387  |
| Tot                    | 345 | 654    | 225  | 1224 |

Stacked bar diagrams

The two diagrams show the **same bars**, stacked by **Age** or by **Interest** in healthy eating

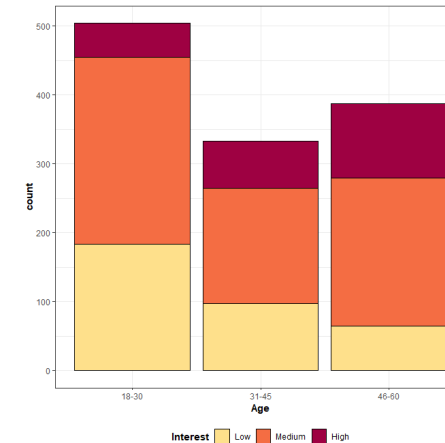
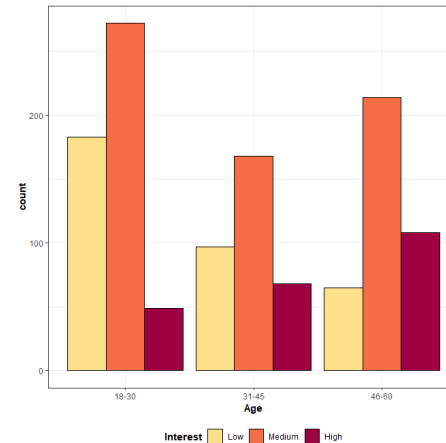
Information is the same provided by side-by-side bars, with more emphasis on the marginal frequencies of the variable on the x-axis (the appearance of the graphs **does not change** when considering relative frequencies)



# Qualitative variables: bar charts

Example: Age and interest in healthy eating (sample of 1224 individuals)

| <i>Interest</i><br><i>Age</i> | Low        | Medium     | High       | Tot         |
|-------------------------------|------------|------------|------------|-------------|
| 18-30                         | 183        | 272        | 49         | <b>504</b>  |
| 31-45                         | 97         | 168        | 68         | <b>333</b>  |
| 46-60                         | 65         | 214        | 108        | <b>387</b>  |
| Tot                           | <b>345</b> | <b>654</b> | <b>225</b> | <b>1224</b> |



The cross-tab and the bar charts used to represent it provide information on the relevance of the observed pairs of values, but **do not always allow an easy comparison of the relevance of the values of one variable for given values of the other.**

For a more accurate analysis, it is important to compare **the conditional** frequency distributions, i.e. the frequency distributions **of one variable** (e.g. **Interest**) for given values of the other, i.e. **in the subsets of cases characterised by the different categories of the other** (in this case, **Age**).

# Qualitative variables: conditional distributions

Example: Age and interest in healthy eating (sample of 1224 individuals)

| <i>Interest</i><br><i>Age</i> | Low | Medium | High | Tot  |
|-------------------------------|-----|--------|------|------|
| 18-30                         | 183 | 272    | 49   | 504  |
| 31-45                         | 97  | 168    | 68   | 333  |
| 46-60                         | 65  | 214    | 108  | 387  |
| Tot                           | 345 | 654    | 225  | 1224 |

## Row conditional distributions

| <i>Interest</i><br><i>Age</i> | Low   | Medium | High  | Tot  |
|-------------------------------|-------|--------|-------|------|
| 18-30                         | 0.363 | 0.540  | 0.097 | 1.00 |
| 31-45                         | 0.291 | 0.505  | 0.204 | 1.00 |
| 46-60                         | 0.168 | 0.553  | 0.279 | 1.00 |

Note a high % of low interest among the youngest and a high % of high interest among seniors

**Row conditional frequency:** frequency of each category of the column variable  $Y$ , in the subset of cases characterised by one of the categories of  $X$ .

E.g. The frequency of **Interest = Medium** conditioned on the category **Age = 31-45** is:

$$\text{Freq \{Interest = Medium \mid Age = 31-45\}} = \frac{168}{333} = 0.505$$

# Qualitative variables: conditional distributions

Example: Age and interest in healthy eating (sample of 1224 individuals)

| <i>Interest</i><br><i>Age</i> | Low | Medium | High | Tot  |
|-------------------------------|-----|--------|------|------|
| 18-30                         | 183 | 272    | 49   | 504  |
| 31-45                         | 97  | 168    | 68   | 333  |
| 46-60                         | 65  | 214    | 108  | 387  |
| Tot                           | 345 | 654    | 225  | 1224 |

**Column** conditional distributions

| <i>Interest</i><br><i>Age</i> | Low   | Medium | High  |
|-------------------------------|-------|--------|-------|
| 18-30                         | 0.530 | 0.416  | 0.218 |
| 31-45                         | 0.281 | 0.257  | 0.302 |
| 46-60                         | 0.188 | 0.327  | 0.480 |
| Tot                           | 1.00  | 1.00   | 1.00  |

Note that among those who are most interested, seniors prevail, whilst among respondents with low/medium interest, youngsters prevail

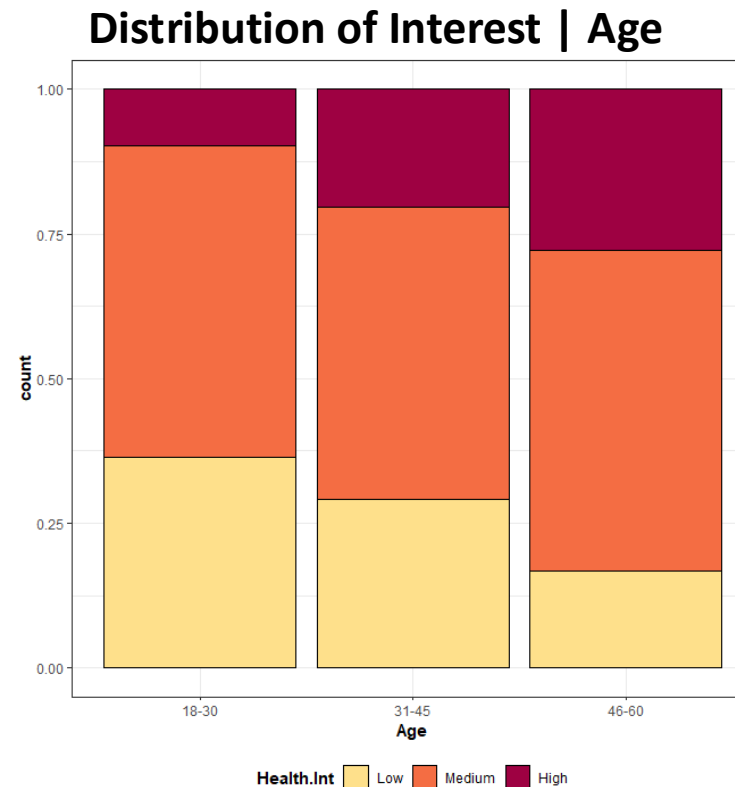
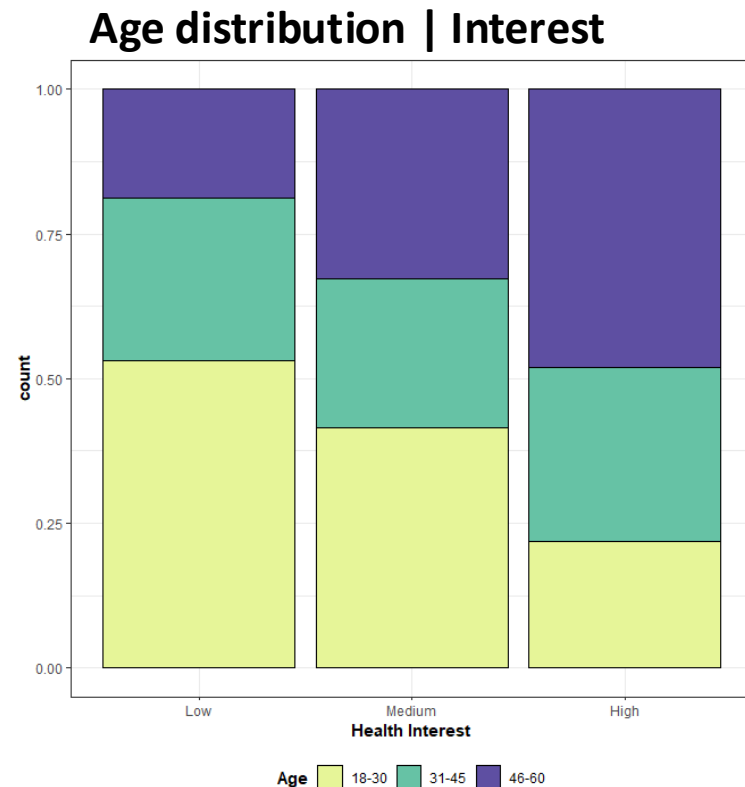
**Column conditional frequency** : frequency of each category of the row variable  $X$ , in the **subset of** cases characterised by one of the categories of  $Y$ .

E.g. The frequency of **Age** = 46-60 **conditioned on** the category **Interest** = *High* is:

$$\text{Freq \{Age = 46-60 | Interest = High\}} = \frac{68}{225} = 0.480$$

# Qualitative variables: conditional distributions

In order to represent conditional distributions, **side-by-side or stacked bar charts** are usually used, displaying conditional rather than joint frequencies.



## Stacked bar diagrams

Note that the height of the bars in the stacked bars diagram is 1, since each bar represents a frequency distribution (sum of frequencies conditional on each category = 1)

# Rstudio: `distr.table.xy()`

The function `distr.table.xy()` allows to tabulate joint or conditional distributions. The syntax is similar to that of `distr.table.x()`. In the simple case of two variables taking few distinct values (qualitative or discrete numeric) and/or factors, the syntax is as follows

```
distr.table.xy(x,y, freq=c("counts"), freq.type=c("joint"), total=T, data)
```

where:

- ▶ **x** and **y** are defined as in `distr.table.x()`: **x** is reported on the **rows** and **y** on the **columns**
- ▶ **freq**: **vector** indicating the frequencies to be reported (a table is produced for each type). Available options are **counts**, **percentages**, **proportions** (possible abbreviations, e.g. **prop**).
- ▶ **freq.type**: **vector** indicating the requested types of frequencies (a table is produced for each type). The options are **joint**, **x|y** (or **column**), and **y|x** (or **row**).
- ▶ **total**: **logical** value specifying whether totals are to be reported in the table or not.
- ▶ **Note**: it is also possible to obtain tables for
  - ▶ **numeric variables to be classified into intervals**, using the **breaks.x** and/or **breaks.y** arguments to specify interval endpoints
  - ▶ **variables measured in classes** by specifying **interval.x=T** and/or **interval.y=T**



# Rstudio: `distr.plot.xy()` - bars

With a similar syntax, the function `distr.plot.xy()` allows to graphically display joint or conditional distributions. To obtain **bar charts**, the syntax is as follows

```
distr.plot.xy(x,y, freq="counts", freq.type="joint",  
              plot.type="bars", bar.type="stacked",data)
```

where (other options similar to `distr.table.x()`):

- ▶ **x** and **y** are defined as in `distr.table.xy()`: Note that in this case **x** is **always on the horizontal axis** and **y** is **always on the vertical axis**
- ▶ **freq, freq.type**: **individual value** indicating which frequencies are to be plotted.
- ▶ **plot.type**: **type of graph** (in this case, **"bars"**; other options will be explained later)
- ▶ **bar.type**: indicates whether a **stacked** (**bar.type = "stacked"**, default) or a side-by-side (**bar.type = "beside"**) bar plot should be built
- ▶ **Note**: plots can also be obtained for
  - ▶ **numeric variables to be classified into intervals**, using the **breaks.x** and/or **breaks.y** arguments to specify interval endpoints
  - ▶ **variables measured in classes by specifying** **interval.x=T** and/or **interval.y=T**

# Hands on: tables&plots for qualitative/discrete variables

Monitoring and evaluation the performance of the call centre of a telecommunication company: survey of a randomly selected sample of users. Results regarding level of satisfaction (**Satisf**) and reason for contact (**Reason**, **Landline**, **Mobile**, **Activ/Transf** - activation/transfer, **Admin** - administrative) are contained in the **CallCenter** dataframe (in the file "**Lesson 8-9\_Data.Rdata**")

## Separate analysis

```
distr.table.x(x=Reason, freq= c("counts","prop"), data=CallCenter)
distr.table.x(x=Satisf, freq= c("counts","prop"), data=CallCenter)
# define a factor to order the levels of variable Satisf
CallCenter$Satisf.F<-factor(CallCenter$Satisf,
                             levels=c("Not at all","Slightly","Moderately",
                                       "Very","Completely"))
distr.table.x(x=Satisf.F, freq= c("counts","prop","cum"), data=CallCenter)
```

# Hands on: tables&plots for qualitative/discrete variables

Monitoring and evaluation of the performance of a call center: level of satisfaction (**Satisf.F**) and reason for contact (**Reason**)

| Satisf.F   | $f_j$ | $p_j$ | $F_j$ |
|------------|-------|-------|-------|
| Not at all | 68    | 0.031 | 0.031 |
| Slightly   | 111   | 0.051 | 0.082 |
| Moderate   | 334   | 0.153 | 0.235 |
| Very       | 675   | 0.308 | 0.543 |
| Completely | 1001  | 0.457 | 1     |
| Total      | 2189  | 1.00  |       |

| Reason       | $f_k$ | $p_k$ |
|--------------|-------|-------|
| Landline     | 597   | 0.273 |
| Mobile       | 999   | 0.456 |
| Activ/Transf | 352   | 0.161 |
| Admin        | 241   | 0.110 |
| Total        | 2189  | 1.000 |

## Univariate distributions: What considerations?

Most of the users contacted the call centre for issues related to the **mobile** line (the mode, with a frequency much higher than the other categories).

As for the satisfaction, frequencies are concentrated on the highest levels. Indeed, the mode is **Completely** and the median is **Very**: most users are completely satisfied and at least 50% are very satisfied.

# Hands on: tables&plots for qualitative/discrete variables

Monitoring and evaluation of the performance of a call center: level of satisfaction (**Satisf.F**) and reason for contact (**Reason**)

| Satisf.F   | $f_j$ | $p_j$ | $F_j$ |
|------------|-------|-------|-------|
| Not at all | 68    | 0.031 | 0.031 |
| Slightly   | 111   | 0.051 | 0.082 |
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| Admin        | 241   | 0.110 |
| Total        | 2189  | 1.000 |

## Univariate distributions: What considerations?

Analysing the two variables separately does not allow to assess whether there is an **association** between satisfaction and reason for contact, and to determine whether there are differences by office, in order to plan possible improvement interventions.

# Hands on: tables&plots for qualitative/discrete variables

Monitoring and evaluation of the performance of a call center: level of satisfaction (**Satisf.F**) and reason for contact (**Reason**)

Cross-tab and side-by-side barplot

```
distr.table.xy(x=Reason,y=Satisf.F, freq=c("counts", "prop"),
               freq.type="joint", f.digits=3,data=CallCenter)
distr.plot.xy(x=Reason,y=Satisf.F, freq="prop", freq.type="joint",
               plot.type="bars",
               colour=c("red", "orange", "yellow", "green", "darkgreen"),
               data=CallCenter)
```

# Hands on: tables&plots for qualitative/discrete variables

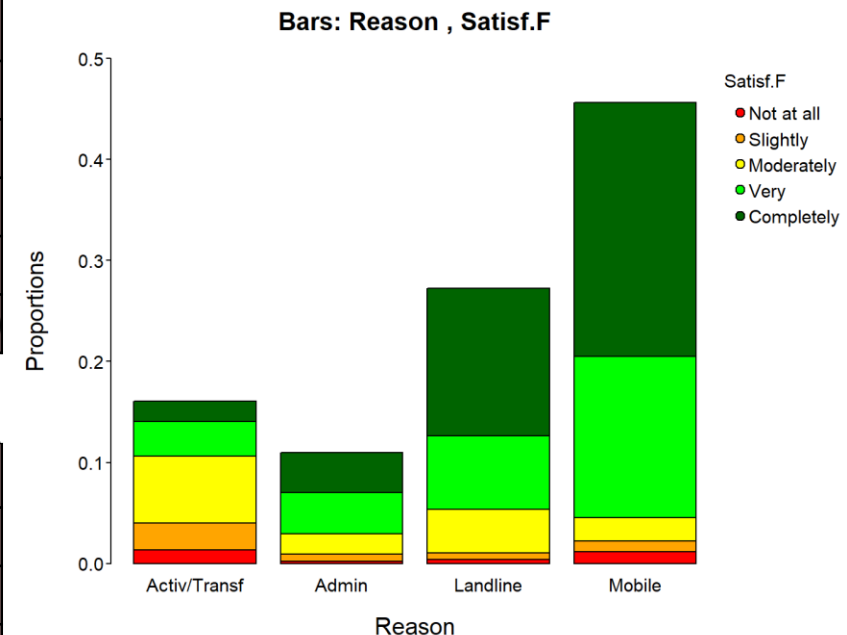
Monitoring and evaluation of the performance of a call center: level of satisfaction (**Satisf.F**) and reason for contact (**Reason**)

**Counts**

| Reason       | Satisf.F   |          |            |      |            | TOTAL |
|--------------|------------|----------|------------|------|------------|-------|
|              | Not at all | Slightly | Moderately | Very | Completely |       |
| Activ/Transf | 29         | 58       | 146        | 75   | 44         | 352   |
| Admin        | 5          | 15       | 44         | 90   | 87         | 241   |
| Landline     | 8          | 15       | 94         | 160  | 320        | 597   |
| Mobile       | 26         | 23       | 50         | 350  | 550        | 999   |
| TOTAL        | 68         | 111      | 334        | 675  | 1001       | 2189  |

**Prop**

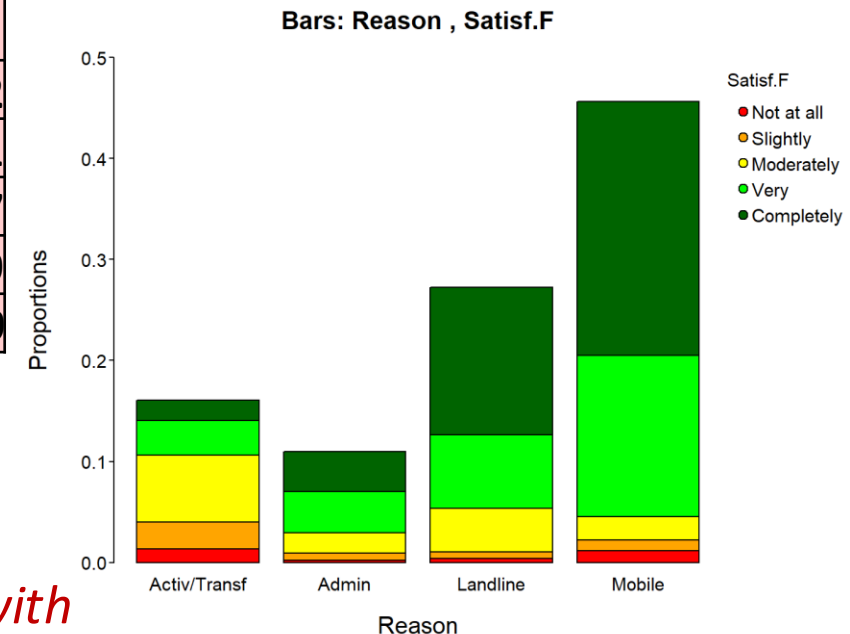
| Reason       | Satisf.F   |          |            |       |            | TOTAL |
|--------------|------------|----------|------------|-------|------------|-------|
|              | Not at all | Slightly | Moderately | Very  | Completely |       |
| Activ/Transf | 0.013      | 0.026    | 0.067      | 0.034 | 0.02       | 0.161 |
| Admin        | 0.002      | 0.007    | 0.02       | 0.041 | 0.04       | 0.11  |
| Landline     | 0.004      | 0.007    | 0.043      | 0.073 | 0.146      | 0.273 |
| Mobile       | 0.012      | 0.011    | 0.023      | 0.16  | 0.251      | 0.456 |
| TOTAL        | 0.031      | 0.051    | 0.153      | 0.308 | 0.457      | 1     |



# Hands on: tables&plots for qualitative/discrete variables

Monitoring and evaluation of the performance of a call center: level of satisfaction (**Satisf.F**) and reason for contact (**Reason**)

| Counts<br>Reason | Satisf.F   |          |            |      |            | TOTAL |
|------------------|------------|----------|------------|------|------------|-------|
|                  | Not at all | Slightly | Moderately | Very | Completely |       |
| Activ/Transf     | 29         | 58       | 146        | 75   | 44         | 352   |
| Admin            | 5          | 15       | 44         | 90   | 87         | 241   |
| Landline         | 8          | 15       | 94         | 160  | 320        | 597   |
| Mobile           | 26         | 23       | 50         | 350  | 550        | 999   |
| TOTAL            | 68         | 111      | 334        | 675  | 1001       | 2189  |



*The joint distribution allows to identify the most relevant combinations of categories.*

*However, to study in depth the level of satisfaction of users with different issues, one must consider **conditional distributions**, providing information about the relevance of the satisfaction levels **within each segment**.*

# Hands on: tables&plots for qualitative/discrete variables

Monitoring and evaluation of the performance of a call center: level of satisfaction (**Satisf.F**) and reason for contact (**Reason**)

Conditional distribution:

```
# conditional distributions of Satisf (y) | Reason (x, horizontal axis)
distr.table.xy(x=Reason,y=Satisf.F, freq="prop", freq.type="y|x",
               data=CallCenter)
distr.plot.xy(x=Reason,y=Satisf.F, freq="prop", freq.type="y|x",
              plot.type="bars",
              colour=c("red", "orange", "yellow", "green", "darkgreen"),
              data=CallCenter)
```



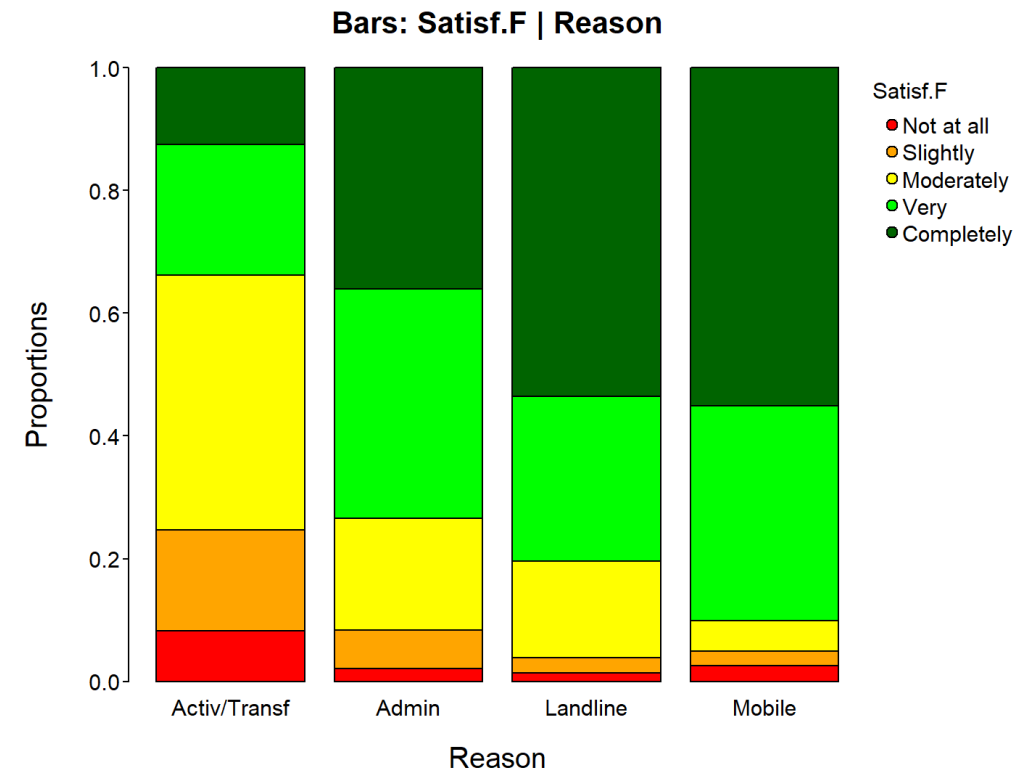
# Hands on: tables&plots for qualitative/discrete variables

Monitoring and evaluation of the performance of a call center: level of satisfaction (**Satisf.F**) and reason for contact (**Reason**)

| Reason       | Satisf.F   |          |            |      |            | TOTAL |
|--------------|------------|----------|------------|------|------------|-------|
|              | Not at all | Slightly | Moderately | Very | Completely |       |
| Activ/Transf | 0.08       | 0.16     | 0.41       | 0.21 | 0.12       | 1     |
| Admin        | 0.02       | 0.06     | 0.18       | 0.37 | 0.36       | 1     |
| Landline     | 0.01       | 0.03     | 0.16       | 0.27 | 0.54       | 1     |
| Mobile       | 0.03       | 0.02     | 0.05       | 0.35 | 0.55       | 1     |

**Comments?**

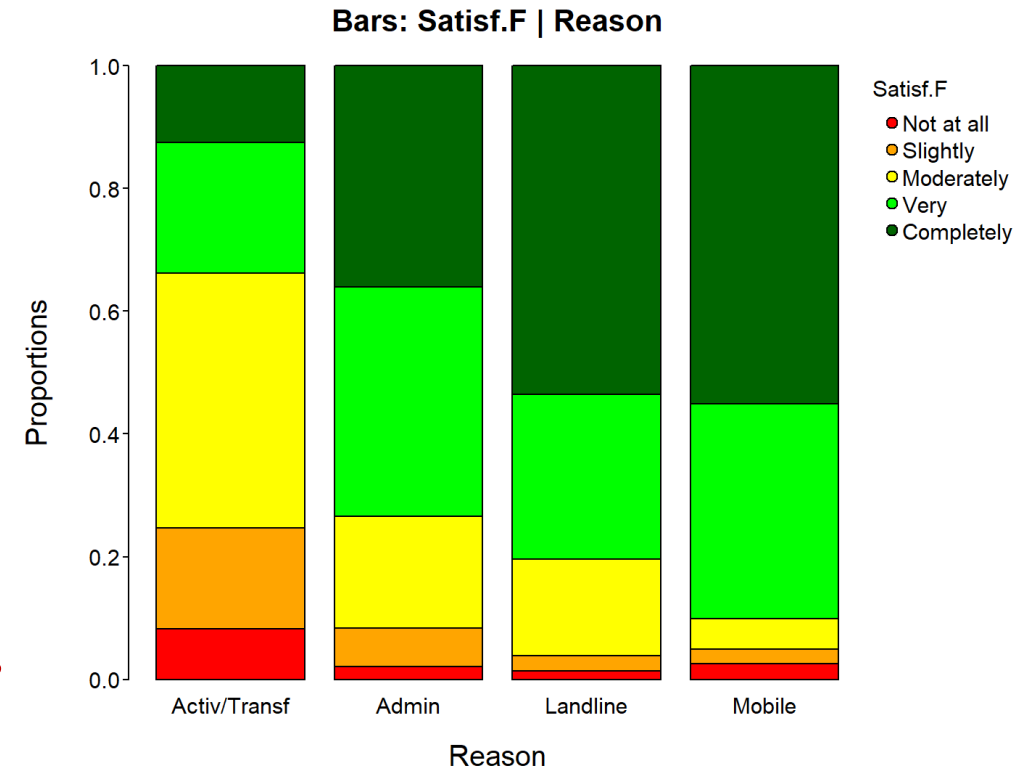
*The conditional distributions are different, and the two variables are associated: the level of satisfaction has a different distribution depending on the reason for contact.*



# Hands on: tables&plots for qualitative/discrete variables

Monitoring and evaluation of the performance of a call center: level of satisfaction (**Satisf.F**) and reason for contact (**Reason**)

| Y X<br>Reason | Satisf.F   |          |            |      |            | TOTAL |
|---------------|------------|----------|------------|------|------------|-------|
|               | Not at all | Slightly | Moderately | Very | Completely |       |
| Activ/Transf  | 0.08       | 0.16     | 0.41       | 0.21 | 0.12       | 1     |
| Admin         | 0.02       | 0.06     | 0.18       | 0.37 | 0.36       | 1     |
| Landline      | 0.01       | 0.03     | 0.16       | 0.27 | 0.54       | 1     |
| Mobile        | 0.03       | 0.02     | 0.05       | 0.35 | 0.55       | 1     |



## Comments?

*The support provided for landline and mobile line-related problems works well, with mode and median satisfaction level equal to 5! For Administration there is a decrease (mode=median=4).*

*On the other hand, the service for Activ/Transf is critical, with mode (definitely dominant) and median both equal to 3, and the highest percentage of dissatisfied or very dissatisfied clients.*

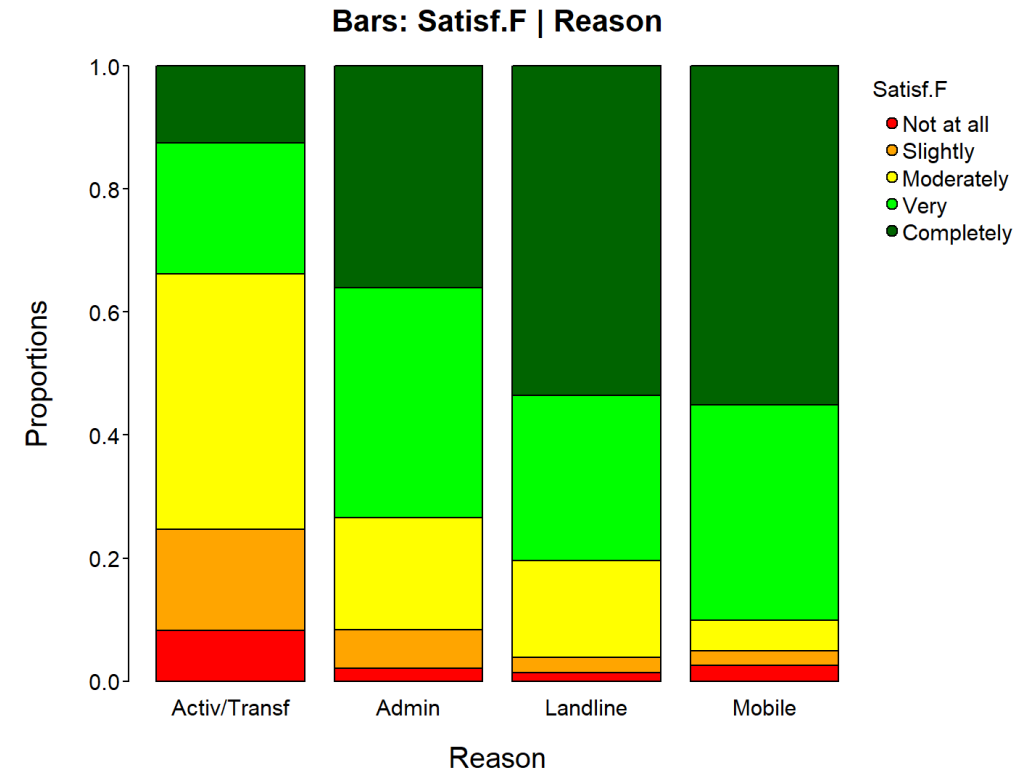
# Hands on: tables&plots for qualitative/discrete variables

Monitoring and evaluation of the performance of a call center: level of satisfaction (**Satisf.F**) and reason for contact (**Reason**)

| Y X<br>Reason | Satisf.F   |          |            |      |            | TOTAL |
|---------------|------------|----------|------------|------|------------|-------|
|               | Not at all | Slightly | Moderately | Very | Completely |       |
| Activ/Transf  | 0.08       | 0.16     | 0.41       | 0.21 | 0.12       | 1     |
| Admin         | 0.02       | 0.06     | 0.18       | 0.37 | 0.36       | 1     |
| Landline      | 0.01       | 0.03     | 0.16       | 0.27 | 0.54       | 1     |
| Mobile        | 0.03       | 0.02     | 0.05       | 0.35 | 0.55       | 1     |

Comments?

*The mode and median of Satisfaction at the marginal level are due to the aggregation of segments of clients having different levels of satisfaction. The apparent good performance at the overall level is due to the high number of customers contacting the call centre for fixed-line and mobile reasons, who are typically very satisfied!!!*



# Hands on: tables&plots for qualitative/discrete variables

Monitoring and evaluation of the performance of a call center: level of satisfaction (**Satisf.F**) and reason for contact (**Reason**)

| Y X<br>Reason | Satisf.F   |          |            |      |            | TOTAL |
|---------------|------------|----------|------------|------|------------|-------|
|               | Not at all | Slightly | Moderately | Very | Completely |       |
| Activ/Transf  | 0.08       | 0.16     | 0.41       | 0.21 | 0.12       | 1     |
| Admin         | 0.02       | 0.06     | 0.18       | 0.37 | 0.36       | 1     |
| Landline      | 0.01       | 0.03     | 0.16       | 0.27 | 0.54       | 1     |
| Mobile        | 0.03       | 0.02     | 0.05       | 0.35 | 0.55       | 1     |

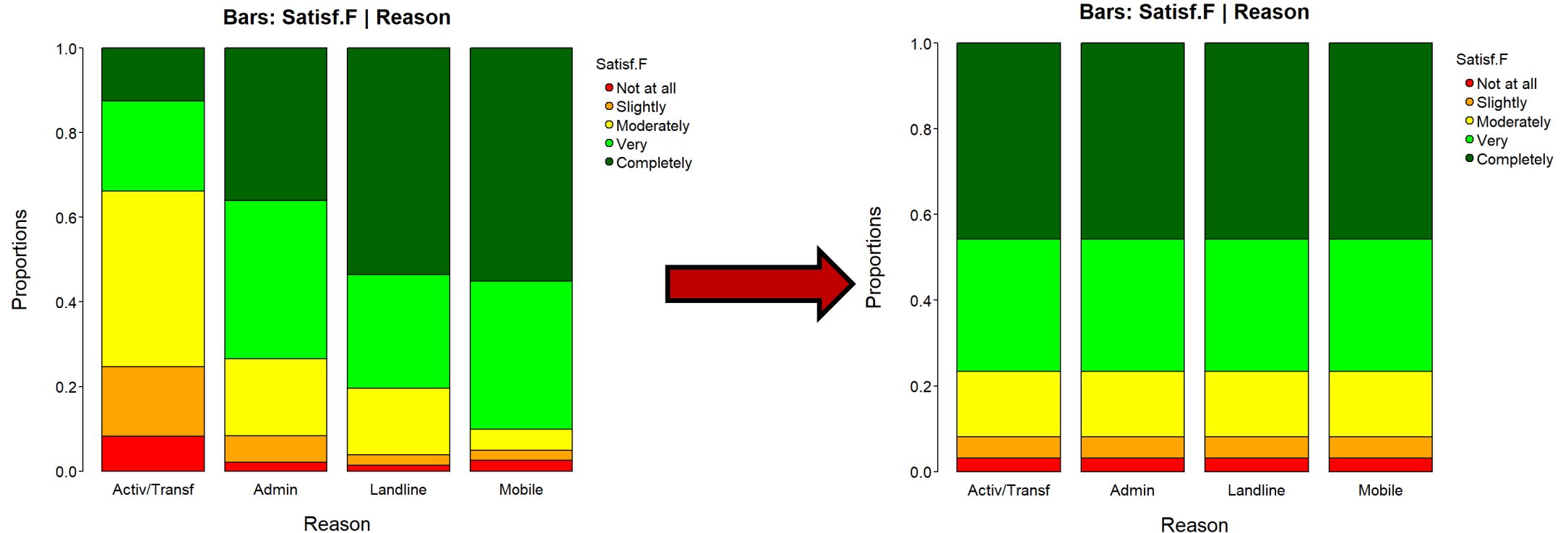
Given the **marginal** distributions, which conditional distributions do you expect in the case of no association (independence)?

They should all coincide with the marginal distribution; this would imply that the level of satisfaction has exactly the same distribution *regardless of the reason for contact.*

| Prop<br>Reason | Satisf.F   |          |            |       |            | TOTAL |
|----------------|------------|----------|------------|-------|------------|-------|
|                | Not at all | Slightly | Moderately | Very  | Completely |       |
| Activ/Transf   | 0.013      | 0.026    | 0.067      | 0.034 | 0.02       | 0.161 |
| Admin          | 0.002      | 0.007    | 0.02       | 0.041 | 0.04       | 0.11  |
| Landline       | 0.004      | 0.007    | 0.043      | 0.073 | 0.146      | 0.273 |
| Mobile         | 0.012      | 0.011    | 0.023      | 0.16  | 0.251      | 0.456 |
| TOTAL          | 0.031      | 0.051    | 0.153      | 0.308 | 0.457      | 1     |

# Hands on: tables&plots for qualitative/discrete variables

Monitoring and evaluation of the performance of a call center: level of satisfaction (**Satisf.F**) and reason for contact (**Reason**)



In the case of *independence*, the distribution of Satisf.F does not change *regardless of the reason of contact: All the conditional distributions coincide with the marginal distribution of Satisf.F*

## **Joint analysis of two variables**

**Variables of which one is continuous numeric**

# Tables and graphs for quantitative variables [READING]

For quantitative (numerical) variables, one must consider that:

- ▶ The number of values assumed by **discrete variables** can be small (e.g. number of visits to a shop in a certain period), as well as quite large (e.g. years of age of a customer, number of transactions with credit card in a year)
- ▶ **Continuous variables** typically take a different value for each observation, and thus the number of observed values is equal to or close to the number of cases

As mentioned, the tables and graphs introduced to jointly analyse qualitative variables can also be used in the case of two discrete variables both taking a small number of distinct values.

We introduce tools for the joint analysis of two variables of which **at least one is continuous numerical** (or discrete with many distinct values)

**Variables of which one is continuous numeric**

**Continuous numeric variable and qualitative/discrete variable**



# Variables: continuous and qualitative

**Response times to users tweets by 5 companies incorporating social media into their marketing strategy.**

**Company:** we consider 5 companies, identified by their activity (**Airlines**, Express **Courier** and Money Remittance service, **Products**, **Mobile** Services, **Tour** Operator).

**Time:** Response time (in seconds) to tweets with questions or requests for assistance/support/information

For each company, response times to 100 randomly selected tweets were measured.

# Variables: continuous and qualitative - frequency table

Response times to tweets measured on 5 companies.

| Time | Company  |         |          |         |      | Tot |
|------|----------|---------|----------|---------|------|-----|
|      | Airplane | Courier | Products | Telecom | Tour |     |
| 713  | 0        | 1       | 0        | 0       | 0    | 1   |
| 728  | 0        | 1       | 0        | 0       | 0    | 1   |
| 780  | 0        | 2       | 0        | 0       | 0    | 2   |
| 795  | 0        | 1       | 0        | 0       | 0    | 1   |
| 840  | 0        | 0       | 0        | 0       | 1    | 1   |
| 855  | 0        | 1       | 0        | 0       | 0    | 1   |
| 870  | 0        | 1       | 0        | 0       | 0    | 1   |
| 878  | 0        | 1       | 0        | 0       | 0    | 1   |
| 893  | 0        | 1       | 0        | 0       | 0    | 1   |
| 900  | 0        | 1       | 0        | 0       | 0    | 1   |
| 908  | 0        | 2       | 0        | 1       | 2    | 5   |
| ...  | ...      | ...     | ...      | ...     | ...  | ... |
| 3000 | 1        | 0       | 0        | 0       | 0    | 1   |
| 3043 | 1        | 0       | 0        | 0       | 0    | 1   |
| Tot  | 100      | 100     | 100      | 100     | 100  | 500 |

**Cross tabulation** (portion): joint absolute frequencies of the response times and companies

The **continuous** variable **Time** takes (as expected) a very high number of values; the joint frequencies are very low.

The cross-tab is not an effective tool for the analysis of the two variables.

**Which tools for a more efficient analysis?**

# Variables: continuous and qualitative – classifying into intervals

Response times to tweets measured on 5 companies.

| Time        | Company  |         |          |         |      | Tot |
|-------------|----------|---------|----------|---------|------|-----|
|             | Airplane | Courier | Products | Telecom | Tour |     |
| [600,780)   | 0        | 2       | 0        | 0       | 0    | 2   |
| [780,960)   | 0        | 15      | 0        | 11      | 3    | 29  |
| [960,1140)  | 0        | 51      | 16       | 35      | 14   | 116 |
| [1140,1320) | 0        | 27      | 23       | 26      | 34   | 110 |
| [1320,1500) | 0        | 4       | 18       | 21      | 34   | 77  |
| [1500,1680) | 1        | 1       | 14       | 7       | 12   | 35  |
| [1680,1860) | 2        | 0       | 10       | 0       | 3    | 15  |
| [1860,2040) | 6        | 0       | 6        | 0       | 0    | 12  |
| [2040,2220) | 21       | 0       | 7        | 0       | 0    | 28  |
| [2220,2400) | 21       | 0       | 4        | 0       | 0    | 25  |
| [2400,2580) | 27       | 0       | 1        | 0       | 0    | 28  |
| [2580,2760) | 13       | 0       | 0        | 0       | 0    | 13  |
| [2760,2940) | 6        | 0       | 1        | 0       | 0    | 7   |
| [2940,3120] | 3        | 0       | 0        | 0       | 0    | 3   |
| Tot         | 100      | 100     | 100      | 100     | 100  | 500 |

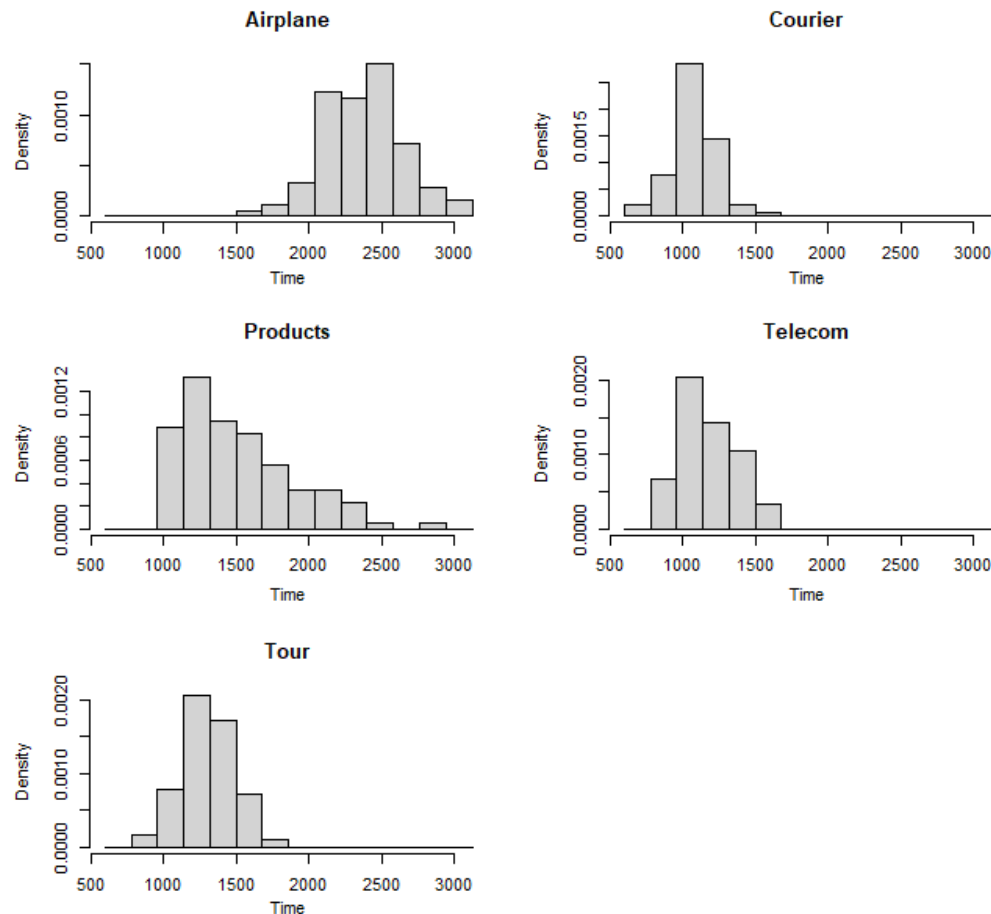
Adopting the same approach followed to analyse a continuous variable, it might seem sensible to classify the **Time** variable into intervals, and to build the cross-tab between time in classes and company.

Equal-width interval classes ( $180 = 3$  minutes)

**Comments?**

# Variables: continuous and qualitative - histograms

## Response times to tweets from 5 companies: histogram comparison

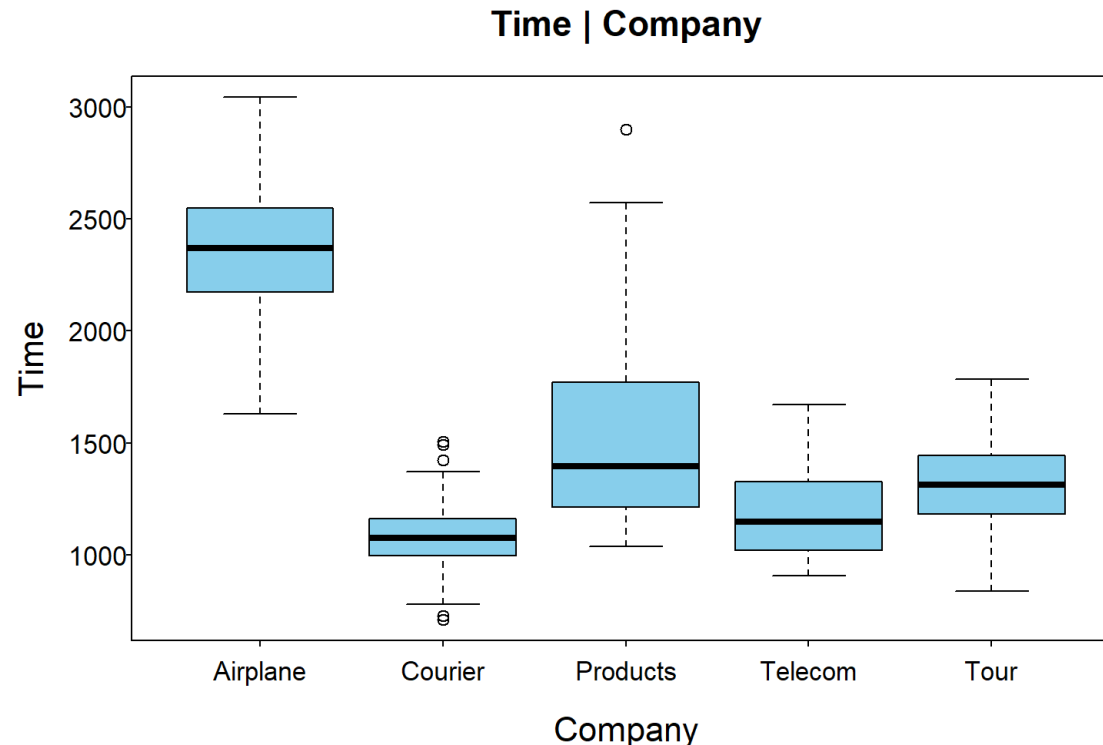


The comparison between response times of different companies is easier if based on histograms built using the same classification for **Time** for every company. Such histograms describe the distribution of the response time **conditional on the company**.

The analysis and comparison of histograms can be more difficult or complicated when the differences between conditional distributions are not marked or when the number of levels of the **conditioning** variable is relatively high.

# Variables: continuous and qualitative - side by side boxplots

Response times to tweets measured on 5 companies: comparison of boxplots.

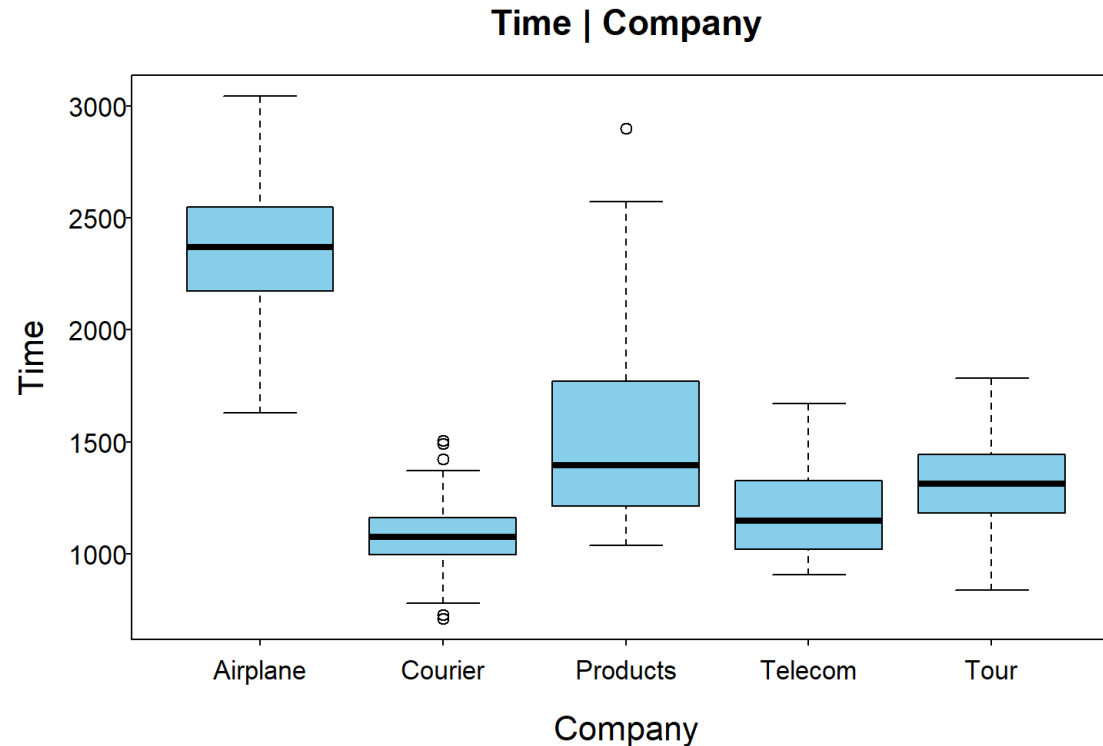


The most effective graphical representation for comparing conditional distributions is based on **side-by-side boxplots**.

Indeed, as already seen for the analysis of a single variable, the boxplot effectively summarises the most relevant features of a variable's distribution, and provides a schematic representation that does not depend on the analyst's choices (concerning for example the choice of the intervals used to build the histograms).

# Variables: continuous and qualitative - boxplots side by side

Response times to tweets measured on 5 companies: comparison of boxplots.



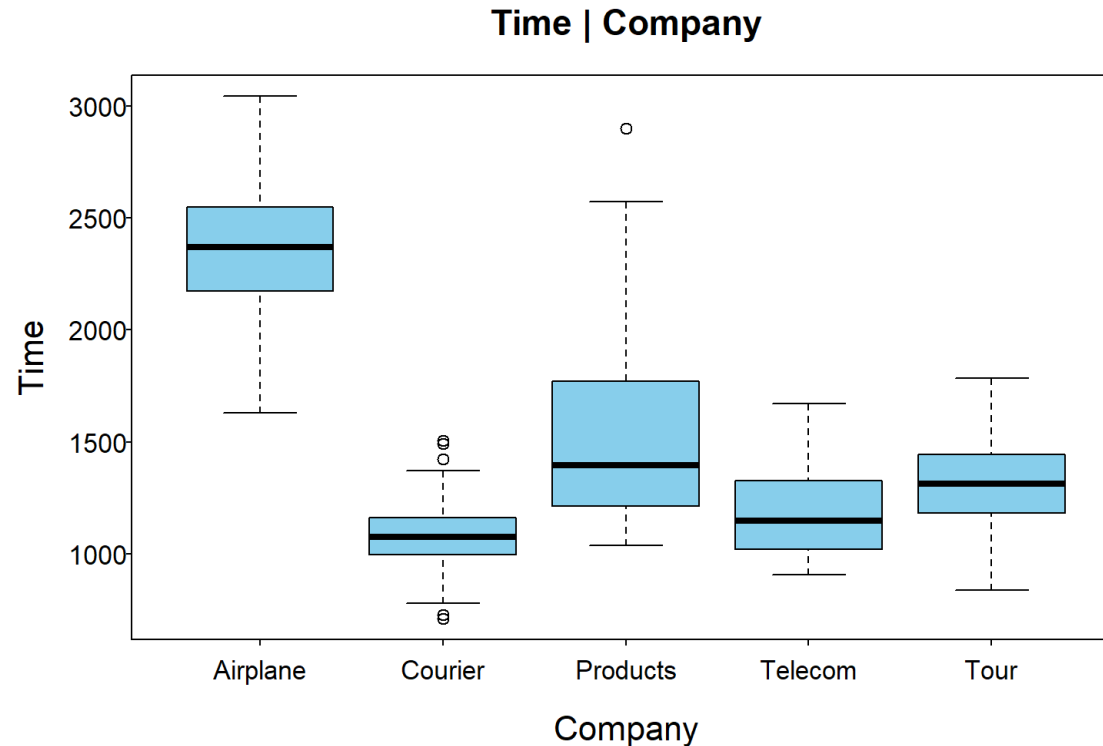
## Comments?

*We note that **Airplane** is the company that “generally” has the slowest response time, followed by **Product**, even if the two distributions of response times for these companies are more dispersed compared to the others.*

*In particular, note that **Airplane** presents a rather symmetric distribution, which partly overlaps with **Product**'s distribution. The latter distribution is skewed to the right; in particular, the median response time - not much higher than that of Courier, Telecom and Tour - is higher than the third quartile of the distributions for Courier and Telecom, and the 25% of the slowest response times are higher than the maximum response times of these 3 companies.*

# Variables: continuous and qualitative - side by side boxplots

Response times to tweets measured on 5 companies: comparison of boxplots.



## Comments?

***Courier** is definitely the fastest responding company in a fairly systematic manner, followed as mentioned by Telecom and Tour.*

*In conclusion, the variable **Time** is associated with the variable **Company**, as its distribution varies as **Company changes**.*

To compare the characteristics of the distributions in a more synthetic way, one can use the summary measures of **Time** conditional on **Company**.

# Variables: continuous and qualitative - summary measures

Response times to tweets measured on 5 companies: conditional summary measures.

Summary measures for Time | Company

Summary measures

| Company  | n   | n.a | min  | q1      | median | mean    | q3      | max  | sd     | var       |
|----------|-----|-----|------|---------|--------|---------|---------|------|--------|-----------|
| Airplane | 100 | 0   | 1629 | 2175.75 | 2370.5 | 2371.95 | 2542.25 | 3043 | 269.69 | 72732.43  |
| Courier  | 100 | 0   | 713  | 998.00  | 1076.5 | 1077.32 | 1163.00 | 1508 | 145.52 | 21176.54  |
| Products | 100 | 0   | 1038 | 1215.75 | 1398.0 | 1519.72 | 1764.25 | 2898 | 397.37 | 157905.78 |
| Telecom  | 100 | 0   | 908  | 1020.00 | 1148.0 | 1185.12 | 1323.75 | 1673 | 186.81 | 34898.53  |
| Tour     | 100 | 0   | 840  | 1187.25 | 1313.0 | 1317.04 | 1442.00 | 1785 | 188.38 | 35485.72  |

Requested statistics

| Company  | n   | n.a | p10    | p90    | p95     |
|----------|-----|-----|--------|--------|---------|
| Airplane | 100 | 0   | 2090.3 | 2695.4 | 2829.45 |
| Courier  | 100 | 0   | 907.2  | 1231.5 | 1299.85 |
| Products | 100 | 0   | 1097.1 | 2143.9 | 2299.70 |
| Telecom  | 100 | 0   | 953.0  | 1448.7 | 1509.50 |
| Tour     | 100 | 0   | 1077.8 | 1583.0 | 1605.40 |



# Rstudio: `distr.plot.xy()` & `summary.x()`

The function `distr.plot.xy()` can be used to easily to obtain **side-by-side boxplots**:

```
distr.plot.xy(x,y, plot.type="boxplot",data)
```

Note that as in the case of the boxplot for a single variable, the plot can only be built for **numeric variables** (classification into intervals is not possible). It is clearly not necessary to indicate the frequencies to be plotted as the boxplot is based on the raw data.

The `distr.summary.x()` function allows to obtain conditional summary measures by specifying, in addition to the arguments already used for the analysis of a single variable, the conditioning variable(s) (no more than 2) using the `by1` and/or `by2` arguments:

```
distr.summary.x(x, by1, by2, stats, digits=2,f.digits=4, data)
```

# Hands on: analysis of tweets

Response times (**Time**) to tweets measured on 5 companies (**Company**): syntax used to obtain the results shown in the slides

```
# original cross-table (INADEQUATE)
distr.table.xy(x=Time,y=Company,freq="counts",freq.type="joint",
               date=Tweets)

# cross-table with Time classified in intervals
distr.table.xy(x=Time,y=Company,freq="counts",freq.type="joint",
               breaks.x=seq(600,3120,by=180),data=Tweets)

# the syntax to obtain a collection of histograms is not reported
# Side-by-side boxplots
distr.plot.xy(x=Company,y=Time,plot.type="boxplot", data=Tweets)
# conditional summary measures (summary reports a collection of stats)
distr.summary.x(x=Time,by1=Company,stats=c("summary",
                                             "p5","p10","p90","p95"),data=Tweets)
```

**Variables of which one is continuous numeric**

**Two numeric variables**

# Variables: numeric (typically continuous)

Data on house prices in a US city.

**price** House price (dollars)

**lotSize** Lot size (square feet )

**age** Age of the house (years)

**landValue** Land value (dollars)

**livingArea** size of the living area (square feet)

**pctCollege** % of individuals in the neighbourhood who a college degree

**bedrooms** Number of bedrooms

**fireplaces** Number of fireplaces

**bathrooms** Number of bathrooms (fraction in the case of bathrooms without shower/tub)

**rooms** Number of rooms

**heating** Heating system

**fuel** Fuel used for heating

**sewer** Sewer system type

**waterfront** Indicates whether the property is on the waterfront

**newConstruction** Indicates whether the property is a new building

**centralAir** Indicates whether the house has air conditioning

# Variables: numeric

## House price data in a US city

|    | price  | lotSize | age | landValue | livingArea | pctCollege | bedrooms | fireplaces | bathrooms |
|----|--------|---------|-----|-----------|------------|------------|----------|------------|-----------|
| 1  | 132500 | 3920.4  | 42  | 50000     | 906        | 35         | 2        | 1          | 1.0       |
| 2  | 181115 | 40075.2 | 0   | 22300     | 1953       | 51         | 3        | 0          | 2.5       |
| 3  | 109000 | 8276.4  | 133 | 7300      | 1944       | 51         | 4        | 1          | 1.0       |
| 4  | 155000 | 17859.6 | 13  | 18700     | 1944       | 51         | 3        | 1          | 1.5       |
| 5  | 86060  | 4791.6  | 0   | 15000     | 840        | 51         | 2        | 0          | 1.0       |
| 6  | 120000 | 29620.8 | 31  | 14000     | 1152       | 22         | 4        | 1          | 1.0       |
| 7  | 153000 | 17424.0 | 33  | 23300     | 2752       | 51         | 4        | 1          | 1.5       |
| 8  | 170000 | 52707.6 | 23  | 14600     | 1662       | 35         | 4        | 1          | 1.5       |
| 9  | 90000  | 36154.8 | 36  | 22200     | 1632       | 51         | 3        | 0          | 1.5       |
| 10 | 122900 | 84506.4 | 4   | 21200     | 1416       | 44         | 3        | 0          | 1.5       |
| 11 | 120000 | 40075.2 | 1   | 22300     | 1624       | 51         | 3        | 0          | 2.0       |
| 12 | 97000  | 4791.6  | 153 | 3100      | 1383       | 57         | 3        | 0          | 2.0       |
| 13 | 127000 | 6098.4  | 9   | 300       | 1300       | 41         | 3        | 0          | 1.5       |
| 14 | 89900  | 0.0     | 88  | 2500      | 936        | 57         | 3        | 0          | 1.0       |
| 15 | 155000 | 5662.8  | 9   | 300       | 1300       | 41         | 3        | 0          | 1.5       |

Joint distribution analysis of **price** and **landValue**? And of **price** and **livingArea** or **rooms**?

Variables are continuous. It is possible to construct a cross table after classification into intervals, but this would imply an over-**simplification** and compression of data. To analyse the **pairs of values actually observed** in more detail, a **scatterplot** is typically used.

# Scatterplot

**Scatterplot:** a graph in which the observed data on two variables are plotted on a Cartesian plane. Each observation is identified by a point in the plane whose coordinates are the values observed on the two variables on the horizontal and vertical axis

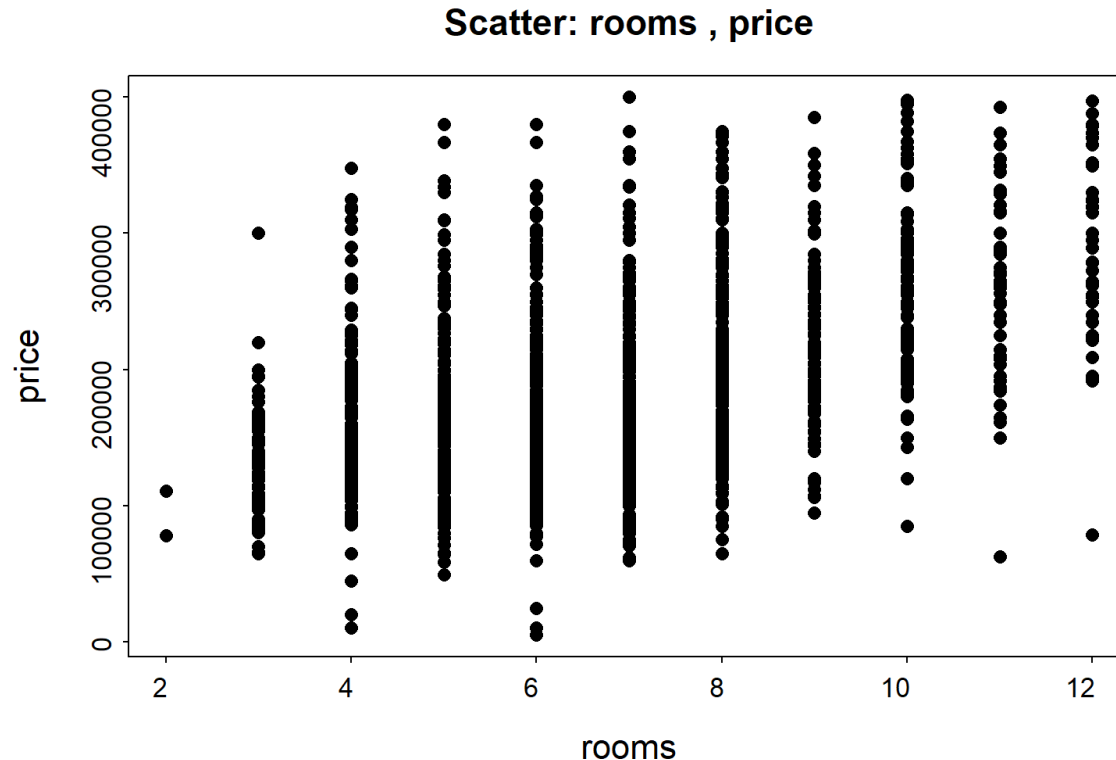


The scatterplot allows to

- ▶ **visualise** the joint distribution
- ▶ identify **any relationships between variables**  
*House prices are positively associated with the size of the living area although there is some dispersion*
- ▶ identify **any outliers**, i.e. points which deviate from the cloud of points  
*Some houses with large living areas have relatively low prices, and some houses have high prices despite having small living areas*

# Scatterplot

## House price data in a US city: prices and number of rooms



One of the two variables is discrete. However, since **price** is continuous, it is better to use the scatterplot, as each observation (house) is characterised by a different **pair** of values on the two variables and the values are both numbers.

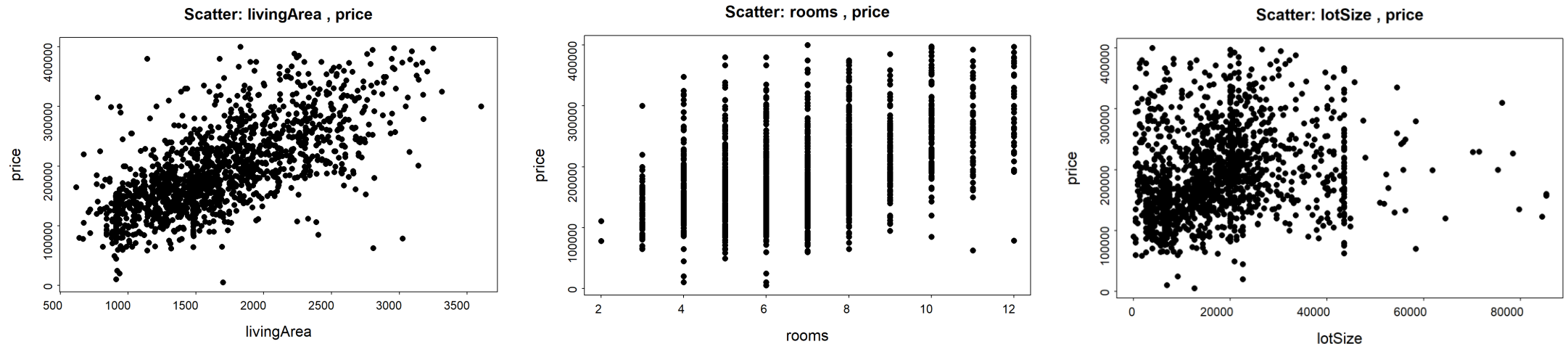
*In this case we observe a positive, but weaker, relationship between the two variables: houses with the same number of rooms have very different prices.*

A comparison of this case with the previous one suggests the need of measures to assess the **strength** of the relationship between variables.

# Analysis of the relationship between numerical variables

In the joint study of two **numerical variables**, we introduced the scatterplot as a graphical tool to describe the relationship between the variables.

► **Example.** Price, living area, number of rooms, lot size for houses in a US city



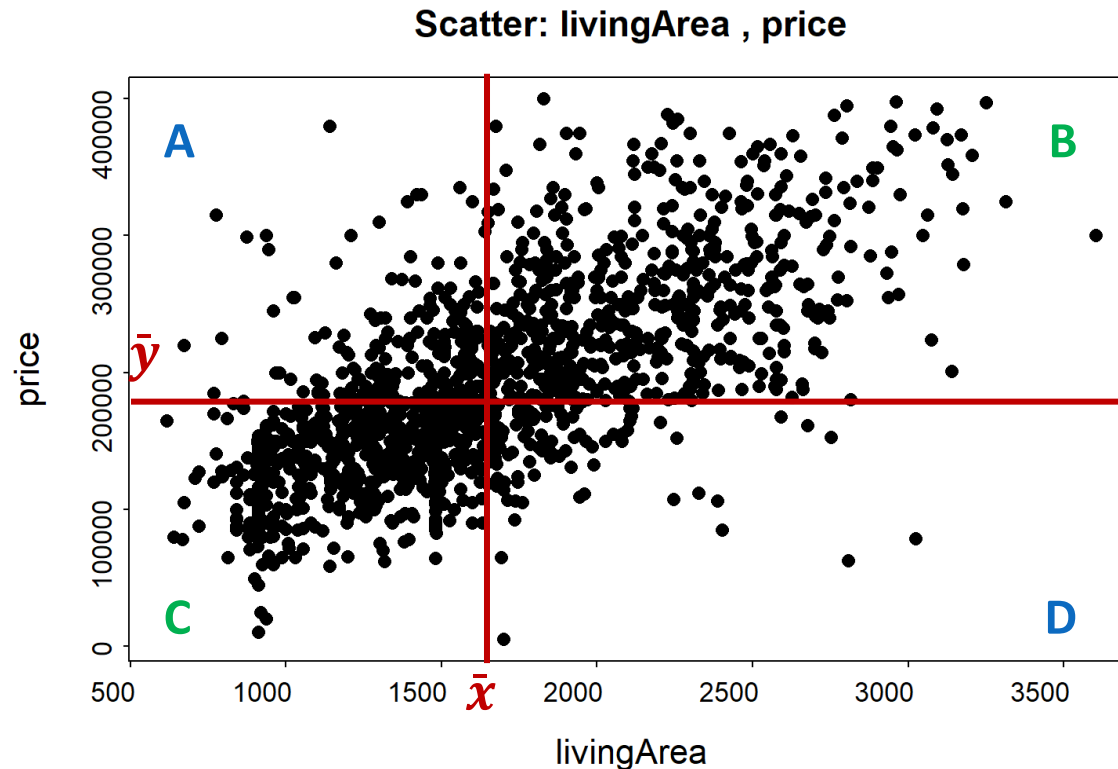
There is a relationship between the three pairs of variables: as the width of the living area, the number of rooms and the size of the property increase, the price of houses increases, but the intensity of the relationship is different!!!

**How to measure and compare the strength of these relationships?**



# Concordant and discordant value pairs

## House price data in a US city: prices and livingArea



Based on the **means of the two variables**, we can identify four quadrants in the plot.  
Quadrants **B** and **C**: **pairs of concordant values**: cases with values both greater or both less than the mean.

Quadrants **A** and **D**: **pairs of discordant values**: cases with values greater than (resp. less than) the mean on one variable and lower than (resp. greater than) the mean on the other variable.

Two variables are **positively (or directly)** associated if concordant pairs prevail and **negatively (or inversely)** associated if discordant pairs prevail.

# Cross-products (cross-products)

To assess the strength of the relation between two variables, we have to consider both the **direction** of the deviations from the means and their **size**. At this aim, we consider for each case  $(x_i, y_i)$  its cross-product, i.e. the product of the deviations from the means

$$(x_i - \bar{x})(y_i - \bar{y})$$

- ▶ The cross-product informs both on the concordance/discordance (sign) of the values observed on the two variables and on the extent of the deviations from the means.
  - ▶ Cases with a **positive** cross-product are **concordant**: they have both values greater or lower than the means (Panels B and C)
  - ▶ Cases with **negative** cross-product are **discordant**: the deviations from the means have opposite signs; for one variable the value is above the mean, for the other it is below the mean (Panels A and D).
- ▶ To assess whether concordant or discordant pairs prevail, i.e. if between two variables a direct or inverse (or no) relation exists, the average of the cross-products is considered, the so-called **covariance**

# Covariance: definition for population and for sample

The definition of the covariance is slightly different for a population and for a sample

- ▶ For data measured on the entire **population** the covariance is defined as:

$$\sigma_{XY} = \frac{1}{N} \sum_{i=1}^N (x_i - \mu_X)(y_i - \mu_Y) \quad \text{where } \mu_X, \mu_Y = \text{means of } X \text{ and } Y \text{ in the population}$$

- ▶ The **sample covariance** instead is defined as:

$$s_{XY} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) \quad \text{where } \bar{x}, \bar{y} = \text{means of } X \text{ e } Y \text{ in the sample}$$

As seen for the variance, this is not exactly the **average of** the product of the deviations, because the sum is divided by **(n - 1)** and not by **n**. The reason for this difference will be explained when discussing about inference, and specifically about estimation

# The covariance: indirect calculation (shortcut) formula

Focusing on the **sample** covariance, we introduce the so-called indirect calculation formula, that can be useful to calculate the covariance in some cases:

$$s_{xy} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \frac{n}{n-1} \left( \frac{1}{n} \sum_{i=1}^n x_i y_i - \bar{x} \bar{y} \right)$$

*The covariance can be calculated based on the mean of the products between values and on the product of the means!*

We will use this formula only in particular cases, and specifically when the covariance is calculated based on aggregated data

# The covariance: indirect calculation formula [OPTIONAL]

Focusing on the **sample covariance**, note that:

$$\begin{aligned} s_{xy} &= \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \frac{1}{n-1} \sum_{i=1}^n (x_i y_i - x_i \bar{y} - y_i \bar{x} + \bar{x} \bar{y}) \\ &= \frac{1}{n-1} \left[ \sum_{i=1}^n (x_i y_i) - \bar{y} \sum_{i=1}^n x_i - \bar{x} \sum_{i=1}^n y_i + \sum_{i=1}^n \bar{x} \bar{y} \right] \\ &= \frac{1}{n-1} \left[ \sum_{i=1}^n (x_i y_i) - n \bar{y} \bar{x} - n \bar{x} \bar{y} + n \bar{x} \bar{y} \right] = \frac{\sum_{i=1}^n x_i y_i - n \bar{x} \bar{y}}{n-1} \end{aligned}$$

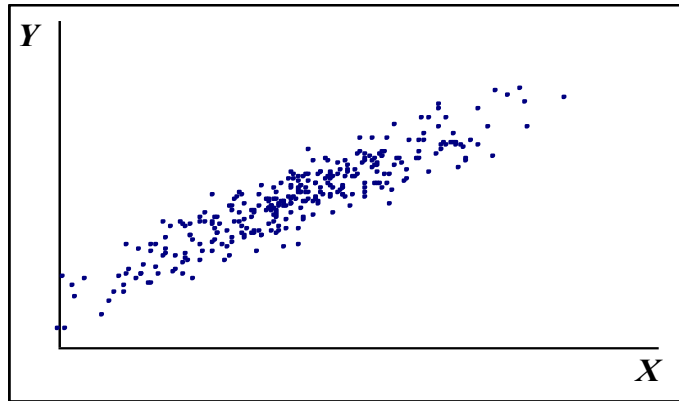
In the case of a **population**, the same reasoning leads to:

$$\sigma_{XY} = \frac{1}{N} \sum_{i=1}^N (x_i - \mu_X)(y_i - \mu_Y) = \frac{\sum_{i=1}^N x_i y_i}{N} - \mu_X \mu_Y$$

# Covariance

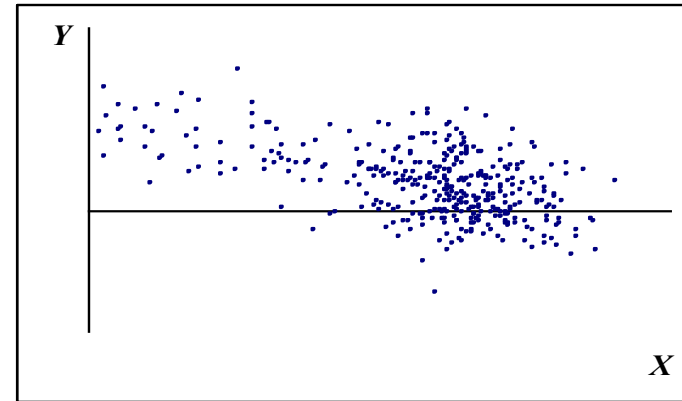
**Positive covariance**

**Concordant pairs prevail**



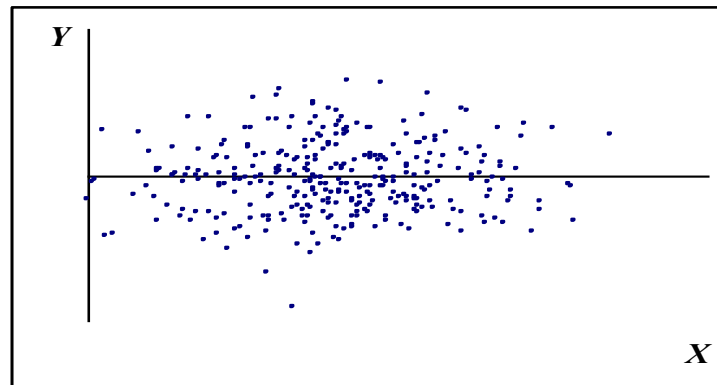
**Negative covariance**

**Discordant pairs prevail**



**Covariance close to zero**

**No prevalence of concordant or discordant pairs is observed**



# Covariance: calculation based on raw data

Example: calculation of the covariance between price and livingArea based on the first  $n = 10$  houses

|       | living Area<br>$x_i$ | price<br>$y_i$ | $(x_i - \bar{x})$ | $(y_i - \bar{y})$ | $(x_i - \bar{x})(y_i - \bar{y})$ |
|-------|----------------------|----------------|-------------------|-------------------|----------------------------------|
|       | 906                  | 132500         | -714.1            | 542.5             | -387399                          |
|       | 1953                 | 181115         | 332.9             | 49157.5           | 16364532                         |
|       | 1944                 | 109000         | 323.9             | -22957.5          | -7435934                         |
|       | 1944                 | 155000         | 323.9             | 23042.5           | 7463466                          |
|       | 840                  | 86060          | -780.1            | -45897.5          | 35804640                         |
|       | 1152                 | 120000         | -468.1            | -11957.5          | 5597306                          |
|       | 2752                 | 153000         | 1131.9            | 21042.5           | 23818006                         |
|       | 1662                 | 170000         | 41.9              | 38042.5           | 1593981                          |
|       | 1632                 | 90000          | 11.9              | -41957.5          | -499294                          |
|       | 1416                 | 122900         | -204.1            | -9057.5           | 1848636                          |
| Somma | 16201                | 1319575        |                   | 0                 | 84167938                         |

The means are

$$\bar{x} = (16201/10) = 1620.1$$

$$\bar{y} = (1319575/10) = 131957.5$$

The sum of the products of the deviations from the means (rounding all intermediate results to 2 decimal places) is **84167938**.

So :

$$s_{xy} = \frac{84167938}{9} = 9351993$$

Comments?

*The covariance is positive: the relationship between the two variables is **direct**.*

# Rstudio: `distr.plot.xy()` & `cov()`

The scatterplot of two variables can be obtained using the function `distr.plot.xy()` :

```
distr.plot.xy(x, y, plot.type="scatter", data)
```

Note that it is not necessary to specify the frequencies to report in the plot because the scatterplot is built based on raw data.

The function `cov()` in R returns the covariance between two **numeric variables** using the syntax:

```
cov(x, y)
```

Note that `x`, `y` are the names of two vectors and they *cannot be* the names of the columns in a dataframe

In the case when some data are missing, they can be excluded from computations by specifying that the calculation of the covariance should be based only on cases with complete information on the two variables:

```
cov(x, y, use="complete")
```



# Hands on: relationship between numerical variables

Scatterplot: house **price** and three numerical variables

```
# scatterplots
distr.plot.xy(x=livingArea,y=price,plot.type =
"scatter",data=House_Price)
distr.plot.xy(x=rooms,y=price,plot.type = "scatter",data=House_Price)
distr.plot.xy(x=lotSize,y=price,plot.type = "scatter",data=House_Price)
```

The R function `cov(x,y)` can be used to calculate the covariance

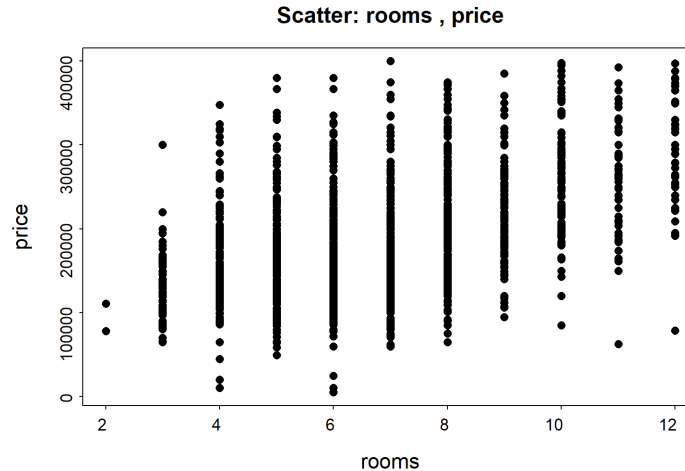
```
> cov(House_Price$livingArea,House_Price$price)
[1] 25617618
> cov(House_Price$rooms,House_Price$price)
[1] 74923.6
> cov(House_Price$lotSize,House_Price$price)
[1] NA
> cov(House_Price$lotSize,House_Price$price,use="complete")
[1] 182595271
```

# Analysis of covariances

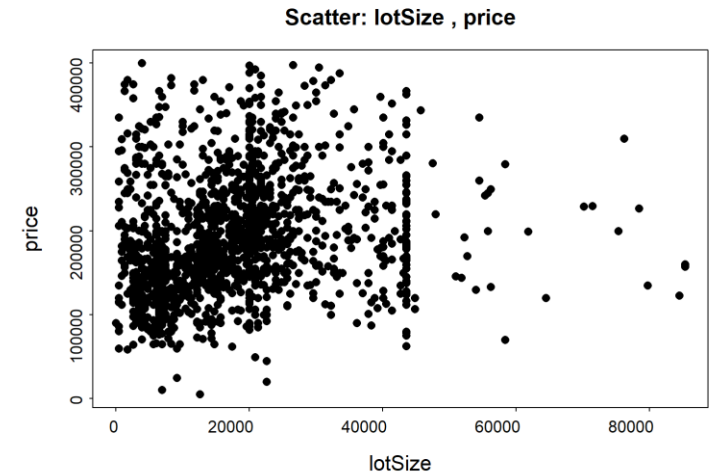
House price data in a US city: Price, living area, number of rooms, property size



**cov = 25617618**



**cov = 74923.6**



**cov = 182595271**

Can we conclude that `price` has the strongest direct relationship with `lotSize`?

*The covariance (as the variance) depends on the **deviations from the means**, and both the means and the deviations depend on the units of measurement of the variables!*

*Thus, covariance allows to deduce the **direction of the relationship between two variables**, but not to assess the strength of such relationship, since it is an **absolute and not a relative measure**!*

# Linear correlation coefficient

A **relative** measure of the **strength of the linear** relationship between two variables is the **linear correlation coefficient**, which is the **ratio of the covariance to the *maximum value it can take***, given by the **product of the mean square deviations of the variables**:

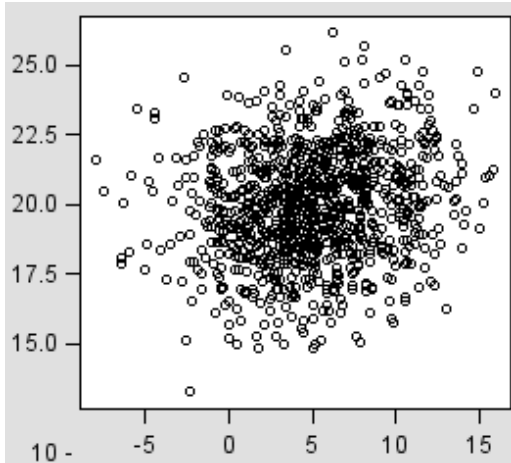
$$r_{xy} = \frac{S_{xy}}{S_x \cdot S_y} \quad [\text{Similar formula for population}].$$

The linear correlation coefficient adjusts the covariance for the unit of measurement, and permits to assess the **strength of the linear relationship between two variables**, that is the extent to which the pairs of observed values tend to **cluster around a straight line**

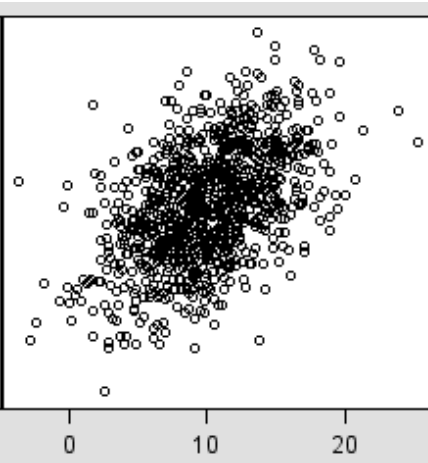
- ▶ It is a **relative measure**, taking values between  $-1$  and  $1$ .
- ▶  $|r_{xy}| = 1$  if and only if the relationship between two variables is **perfectly linear**, i.e. when all the data lie on a line with positive slope ( $r_{xy} = 1$ , direct linear relationship) or negative slope ( $r_{xy} = -1$ , inverse linear relationship).
- ▶  $r_{xy} = 0$  if two variables are not linearly related (**uncorrelated** variables)

# Correlation coefficient

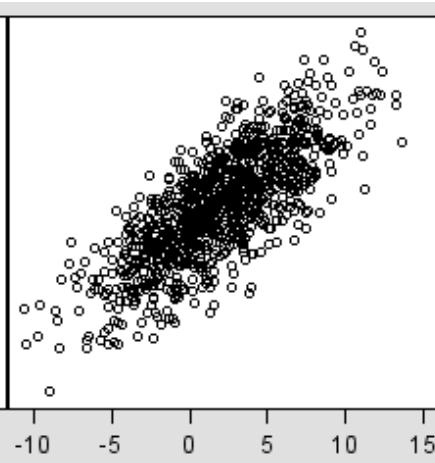
$$r_{xy} = 0.16$$



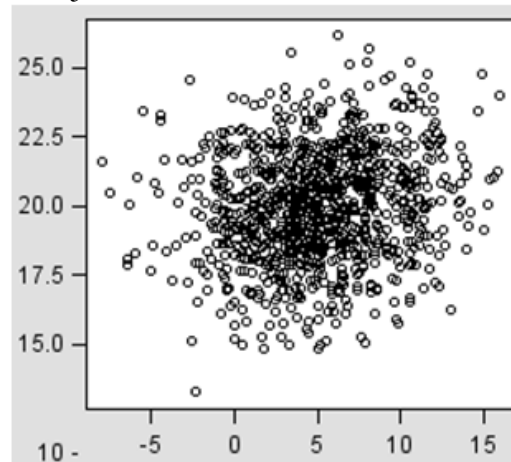
$$r_{xy} = 0.49$$



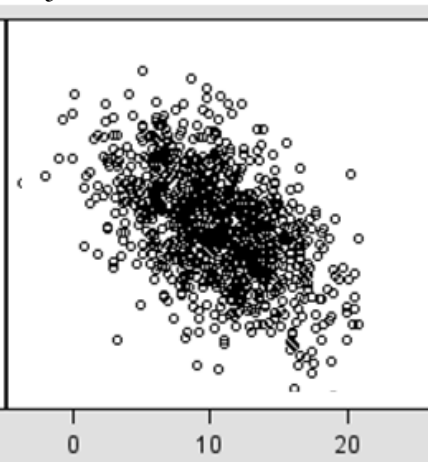
$$r_{xy} = 0.78$$



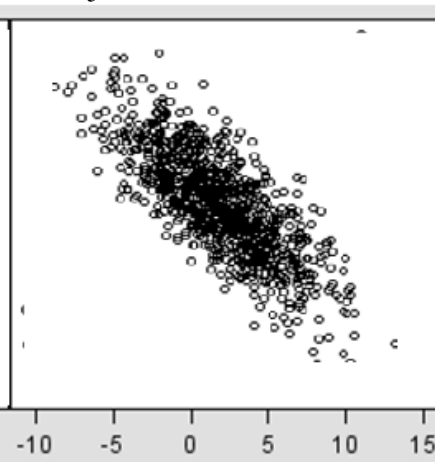
$$r_{xy} = 0.16$$



$$r_{xy} = -0.55$$



$$r_{xy} = -0.86$$

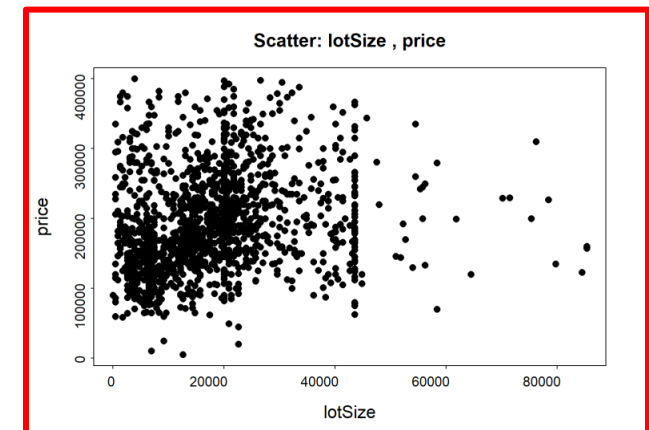
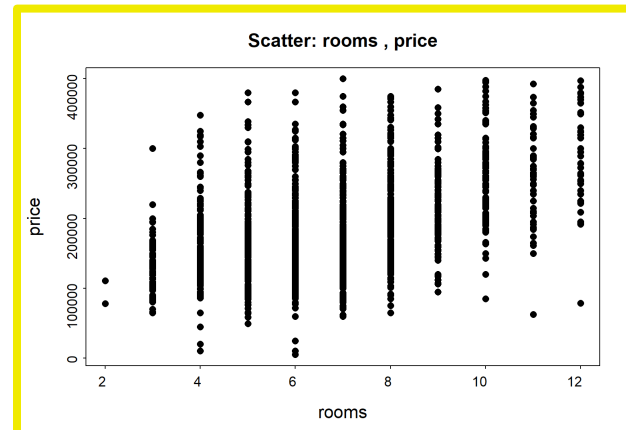
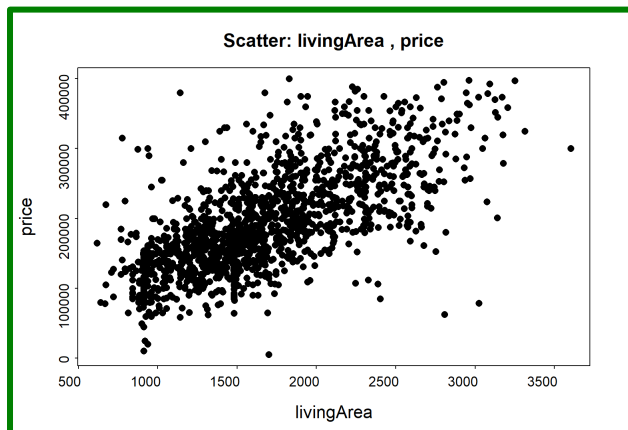


As  $r_{xy}$  moves away from 0 (towards 1 or towards  $-1$ ) the points tend to be more and more concentrated around a straight line (with positive or negative slope depending on the sign of  $r_{xy}$ )

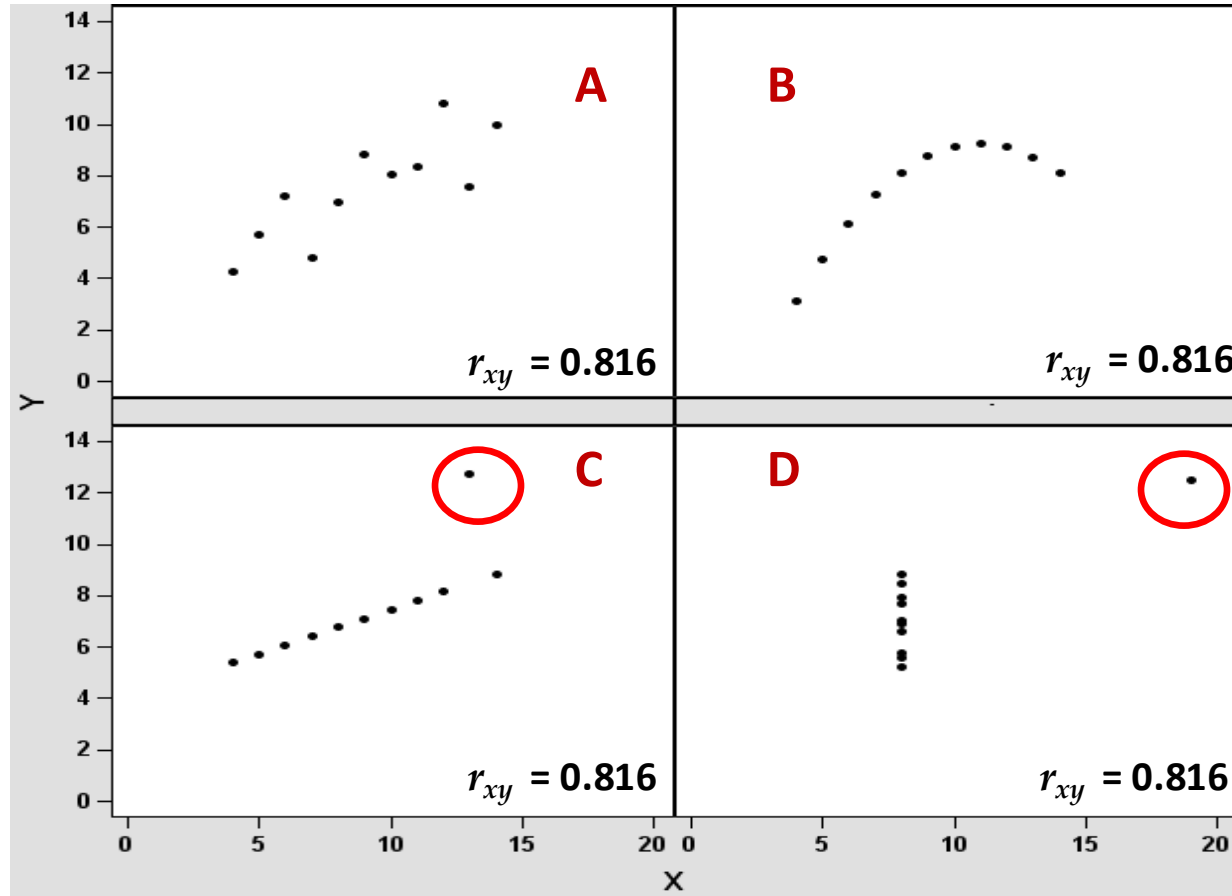
# Hands on: relationship between numerical variables

To calculate the covariance, the R function `cor(x, y)` (very similar to `cor(x, y)`) can be used

```
> cor(House_Price$livingArea, House_Price$price)
[1] 0.685806
> cor(House_Price$rooms, House_Price$price)
[1] 0.4855333
> # to deal with missing values use="complete"
> cor(House_Price$lotSize, House_Price$price, use="complete")
[1] 0.2062668
```



# Correlation: some considerations



*It is important to consider that the correlation in some cases may not be reliable.*

In the 4 cases shown in the plots (Anscombe, 1973), the variables have the same means, variances, covariance and correlation.

However, the relationship between the variables is very different:

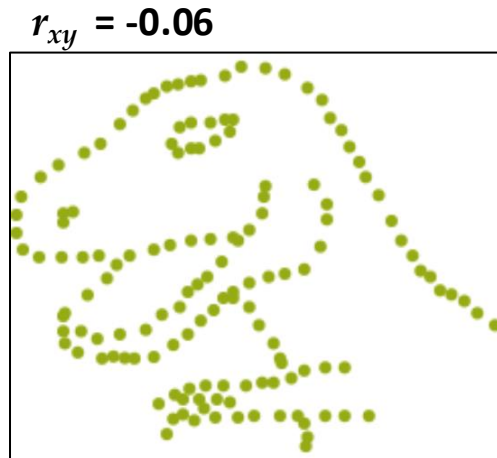
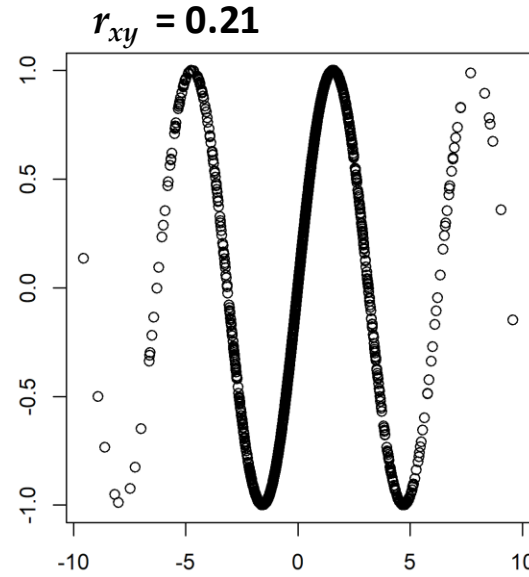
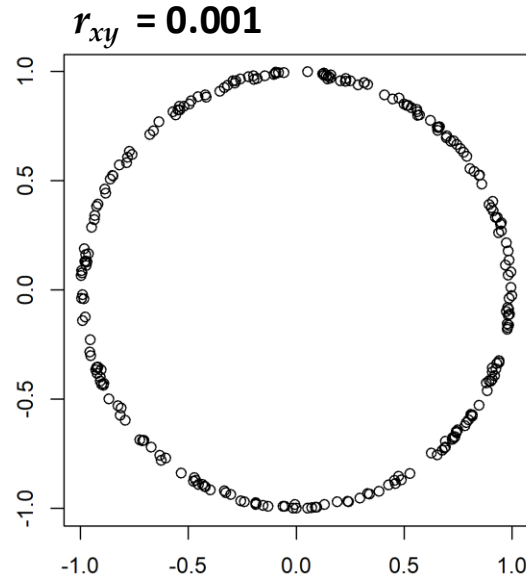
**Plot A:** the correlation properly reflects the relationship between the variables.

**Plot B:** perfect **non-linear** relationship,

**Plot C/D:** the value of  $r_{xy}$  is influenced by **extreme values (outliers)** whose

presence causes a decrease in  $r_{xy}$  (Plot C: without the outlier it would be  $r_{xy}=1$ ) or an increase in  $r_{xy}$  (Plot D: without the outliers it would be  $r_{xy}=0$ )

# Correlation: some considerations



*It is important to avoid forcing the interpretation of the correlation coefficient*

In the plots, the correlations coefficients are very low.

However, it would be totally wrong to conclude that the considered pairs of variables are **not related**.

Actually, the linear correlation coefficient measures the strength of the **linear relationship** and does not allow to draw conclusions on the strength of any non-linear relationship!!!

For other examples:

<https://jumpingrivers.github.io/datasauRus/>

# Conclusions

- ▶ The correlation coefficient can be affected by the presence of outliers.
- ▶ The correlation coefficient may be very low even when there is a relationship between the variables, but the relationship is not linear.
- ▶ At the same time, two variables may be characterized by a medium-high or high correlation coefficient even if their relationship is not linear.

**Analysing the scatterplot** allows assessing whether the correlation coefficient is reliable.